

bargaining games (such as those considered in Appendix A of Chapter 9) by adopting an *axiomatic* point of view. Thus, the approach is more related to ideas of cooperative game theory (as reviewed in Appendix A of Chapter 18).²¹

For current purposes, the description of a *bargaining problem* among I agents is composed of two elements: a *utility possibility* set $U \subset \mathbb{R}^I$ and a *threat*, or *status-quo*, point $u^* \in U$. The set U represents the allocations of utility that can be settled on if there is cooperation among the different agents. The point u^* is the outcome that will occur if there is a breakdown of cooperation. Note that cooperation requires the unanimous participation of all agents, in which case, to repeat, the available utility options are given by $U \subset \mathbb{R}^I$. If one agent does not participate, then the only possible outcome is the vector u^* . This setup is completely general with two agents and, because of this, the two-agent case is our central reference case in this section. With more than two agents, the assumption is a bit extreme, since we may want to allow for the possibility of partial cooperation. We take up this possibility in Section 22.F.

Throughout this section we assume that $U \subset \mathbb{R}^I$ is convex and closed and that it satisfies the free disposal property $U - \mathbb{R}_+^I \subset U$ (i.e. if $u' \leq u$ and $u \in U$ then $u' \in U$). As in Definition 22.B.1, $U \subset \mathbb{R}^I$ could be generated from a set of underlying alternatives X , which could well include lotteries over deterministic outcomes.²² For simplicity we also assume that u^* is interior to U and that $\{u \in U: u \geq u^*\}$ is bounded.

Definition 22.E.1: A *bargaining solution* is a rule that assigns a solution vector $f(U, u^*) \in U$ to every bargaining problem (U, u^*) .²³

We devote the rest of this section to a discussion of some of the properties one may want to impose on $f(\cdot)$ and to a presentation of four examples of bargaining solutions: the *egalitarian*, the *utilitarian*, the *Nash* and the *Kalai-Smorodinsky solutions*. We should emphasize, however, that a strong assumption has already been built into the formalization of our problem: we are implicitly assuming that the solution depends on the set X of feasible alternatives only through the resulting utility values.

Definition 22.E.2: The bargaining solution $f(\cdot)$ is *independent of utility origins* (IUO), or *invariant to independent changes of origins*, if for any $\alpha = (\alpha_1, \dots, \alpha_I) \in \mathbb{R}^I$ we have

$$f_i(U', u^* + \alpha) = f_i(U, u^*) + \alpha_i \quad \text{for every } i$$

whenever $U' = \{(u_1 + \alpha_1, \dots, u_I + \alpha_I): u \in U\}$.

The IUO property says that the bargaining solution does not depend on absolute scales of utility. From now on we assume that this property holds. Note that we therefore always have $f(U, u^*) = f(U - \{u^*\}, 0) + u^*$. This allows us to normalize our problems to $u^* = 0$. From now on we do so and simply write $f(U)$ for $f(U, 0)$.

21. For general introductions to the material of this section, see Roth (1979), Moulin (1988), and Thomson (1995).

22. In principle, the underlying set X and the corresponding utility functions on X could be different for different $U \subset \mathbb{R}^I$. For the theory that follows all that matters is the utility set U .

23. Thus, a bargaining solution is a choice rule in the sense of Chapter 1. If an underlying alternative set X is kept fixed and, therefore, the form of U , generated as in Definition 22.B.1, depends only on the utility functions, we can also regard the bargaining solution as a choice of function in the sense of Definition 21.E.1.

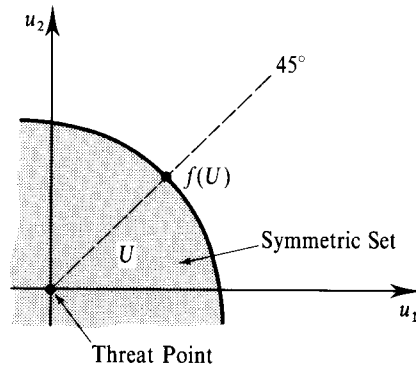


Figure 22.E.1
The symmetry property for bargaining solutions.

It should not be forgotten, however, that a change in the threat point (which will now show up as a change in U) will affect the point settled on.

Definition 22.E.3: The bargaining solution $f(\cdot)$ is *independent of utility units* (IUU), or *invariant to independent changes of units*, if for any $\beta = (\beta_1, \dots, \beta_I) \in \mathbb{R}^I$ with $\beta_i > 0$ for all i , we have

$$f_i(U') = \beta_i f_i(U) \quad \text{for every } i$$

whenever $U' = \{(\beta_1 u_1, \dots, \beta_I u_I) : u \in U\}$.²⁴

With independence of utility origins (implicitly assumed in Definition 22.E.3), independence of utility units tells us that, although the bargaining solution uses cardinal information on preferences, it does not in any way involve interpersonal comparisons of utilities.

Definition 22.E.4: The bargaining solution $f(\cdot)$ satisfies the *Pareto property* (P), or is *Paretian*, if, for every U , $f(U)$ is a (weak) Pareto optimum, that is, there is no $u \in U$ such that $u_i > f_i(U)$ for every i .

Definition 22.E.5: The bargaining solution $f(\cdot)$ satisfies the property of *symmetry* (S) if whenever $U \subset \mathbb{R}^I$ is a symmetric set (i.e., U remains unaltered under permutations of the axes;²⁵ see Figure 22.E.1), we have that all the entries of $f(U)$ are equal.

The interpretation of the symmetry property is straightforward: if, as reflected in U , all agents are identical, then the gains from cooperation are split equally.

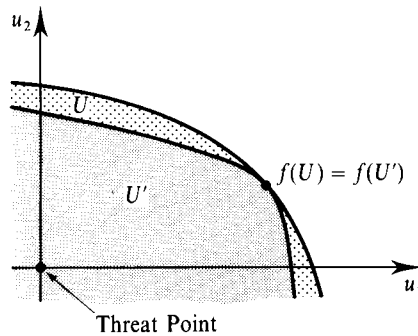
Definition 22.E.6: The bargaining solution $f(\cdot)$ satisfies the property of *individual rationality* (IR) if $f(U) \geq 0$.

In words: the cooperative solution does not give any agent less than the threat point (recall also that, after normalization, we consider only sets U with $0 \in U$). It is a sensible property: if some agent got less than zero, then she would do better by opting out and bringing about the breakdown of negotiation.

The next property is more substantial.

24. Geometrically, U' is obtained from U by stretching the different axes by the rescaling factors $(\beta_1, \dots, \beta_I)$.

25. More precisely, if $u \in U$ then $u' \in U$ for any u' differing from u only by a permutation of its entries.

**Figure 22.E.2**

The property of independence of irrelevant alternatives for bargaining solutions.

Definition 22.E.7: The bargaining solution satisfies the property of *independence of irrelevant alternatives* (IIA) if, whenever $U' \subset U$ and $f(U) \in U'$, it follows that $f(U') = f(U)$.

The IIA condition says that if $f(U)$ is the “reasonable” outcome in U and we consider a U' that is smaller than U but retains the feasibility of $f(U)$, that is, we only eliminate from U “irrelevant alternatives,” then $f(U)$ remains the reasonable outcome (see Figure 22.E.2). This line of justification would be quite persuasive if we could replace “reasonable” by “best.” Indeed, if $f(U)$ has been obtained as the unique maximizer on U of some social welfare function $W(u)$, then the IIA condition is clearly satisfied [if $f(U)$ maximizes $W(\cdot)$ on U then it also maximizes $W(\cdot)$ on $U' \subset U$]. We note that while the converse is not true, it is nonetheless the case that, in practice, the interesting examples where IIA is satisfied involve the maximization of some SWF.

We proceed to present four examples of bargaining solutions. To avoid repetition, we put on record that all of them satisfy the Paretian, symmetry, and individual rationality properties (as well as, by the formulation itself, the independence of utility origins). You are asked to verify this in Exercise 22.E.1. In Exercise 22.E.2 you are asked to construct examples violating some of these conditions.

Example 22.E.1: Egalitarian Solution. At the egalitarian solution $f_e(\cdot)$, the gains from cooperation are split equally among the agents. That is, for every bargaining problem $U \subset \mathbb{R}^I$, $f_e(U)$ is the vector in the frontier of U with all its coordinates equal. Figure 22.E.3 depicts the case $I = 2$. Note also that, as illustrated in the figure, every $f_e(U)$ maximizes the Rawlsian social welfare function $\text{Min} \{u_1, \dots, u_I\}$ on U .

The egalitarian solution satisfies the IIA property (verify this). Clearly, for this solution, utility units are comparable across agents, and so the IUU property is not satisfied.²⁶ ■

Example 22.E.2: Utilitarian Solution. For every U we now let $f_u(U)$ be a maximizer of $\sum_i u_i$ on $U \cap \mathbb{R}_+^I$. If U is strictly convex, then this point is uniquely defined and, therefore, on the domain of strictly convex bargaining problems the IIA property is satisfied. As with the previous example, the solution violates the IUU condition. Figure 22.E.4 illustrates the utilitarian solution in the case $I = 2$. ■

26. Do not forget that the utility values are not absolute values but rather utility differences from the threat point. It is because of this that changes of origins do not matter.

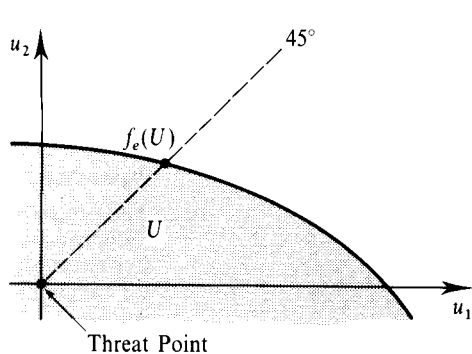


Figure 22.E.3 (left)
The egalitarian solution for bargaining problems.

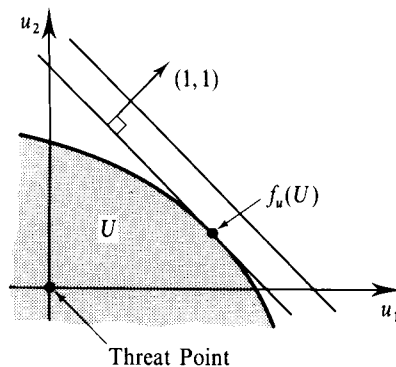


Figure 22.E.4 (right)
The utilitarian solution for bargaining problems.

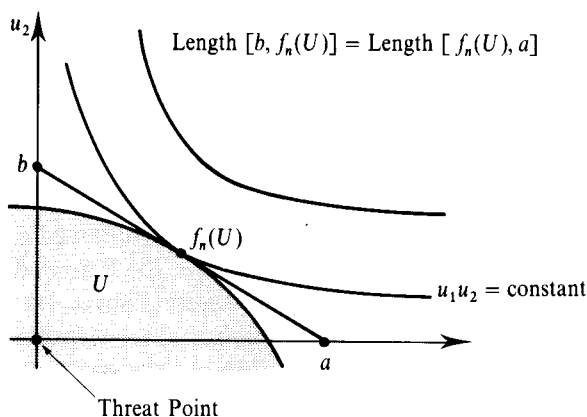
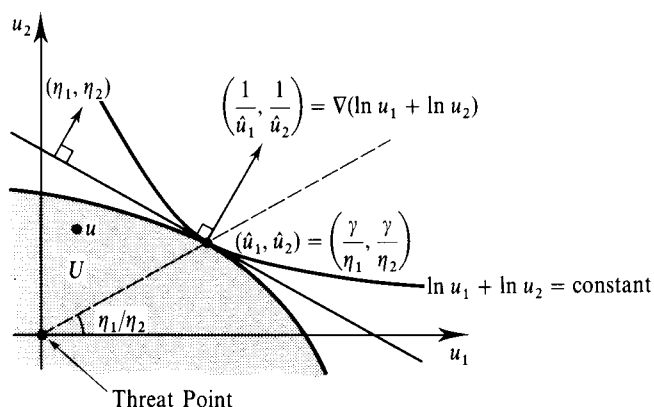


Figure 22.E.5
The Nash solution for bargaining problems.

Example 22.E.3: Nash Solution. For this solution, we take a position intermediate between the two previous examples by requiring that $f_n(U)$ be the point in $U \cap \mathbb{R}_+^I$ that maximizes the product of utilities $u_1 \times \cdots \times u_I$, or, equivalently, that maximizes $\sum_i \ln u_i$ (this corresponds to the case $\rho = 1$ in Example 22.C.4). In Figure 22.E.5, we provide an illustration for $I = 2$. In this case, the Nash solution has a simple geometry: $f_n(U)$ is the boundary point of U through which we can draw a tangent line with the property that its midpoint in the positive orthant is precisely the given boundary point $f_n(U)$; see Exercise 22.E.3.

As with the egalitarian and the utilitarian examples, the Nash solution satisfies the IIA property (because it is defined by the maximization of a strictly concave function). Interestingly, and in contrast to those solutions, *the condition of independence of utility units (IUU) holds for the Nash solution*. To see this, note that $\sum_i \ln u_i \geq \sum_i \ln u'_i$ is equivalent to $\sum_i \ln \beta_i u_i = \sum_i \ln u_i + \sum_i \ln \beta_i \geq \sum_i \ln u'_i + \sum_i \ln \beta_i = \sum_i \ln \beta_i u'_i$ for any constants $\beta_i > 0$. The Nash solution is therefore invariant to whatever origins or units we wish to fix. It depends only on the cardinal characteristics of the utility functions of the agents over the underlying set of alternatives.

There is a way to view the Nash solution as a synthesis of the egalitarian and the utilitarian solutions designed to accomplish the invariance to units: *Given a bargaining problem U , the Nash solution is the only utility outcome that, for some rescaling of units of utility, coincides simultaneously with the utilitarian and the egalitarian solutions.* More formally, suppose that

**Figure 22.E.6**

For some rescaling factors (η_1, η_2) the Nash solution is simultaneously egalitarian and utilitarian.

$\eta_i > 0$ are transformation rates of the given units into new units comparable across agents. If the utilitarian and the egalitarian solutions coincide when applied to the rescaled U' ($= \{(\eta_1 u_1, \dots, \eta_I u_I) : (u_1, \dots, u_I) \in U\}$) then the chosen point $\hat{u} \in U$ must be such that, first, it maximizes $\sum_i \eta_i u_i$ on U and, second, for some $\gamma > 0$ it satisfies $\eta_1 \hat{u}_1 = \dots = \eta_I \hat{u}_I = \gamma$, that is, $\eta_i = \gamma(1/\hat{u}_i)$ for every i . Consider now any $u' \in U$. We have $\sum_i \eta_i u'_i \leq \sum_i \eta_i \hat{u}_i$ and therefore $\sum_i (1/\hat{u}_i) u'_i \leq \sum_i (1/\hat{u}_i) \hat{u}_i$. Since $(1/\hat{u}_1, \dots, 1/\hat{u}_I)$ is the gradient of the concave function $\sum_i \ln u_i$ at $(\hat{u}_1, \dots, \hat{u}_I)$, this implies $\sum_i \ln u'_i \leq \sum_i \ln \hat{u}_i$ (see Section M.C of the Mathematical Appendix). Hence $\hat{u}$ maximizes $\sum_i \ln u_i$ on U , that is, $\hat{u} = f_n(U)$.²⁷ See Figure 22.E.6 for an illustration of the argument. In Exercise 22.E.3 you should show the converse—that the Nash solution is simultaneously utilitarian and egalitarian for appropriate choice of units.

The Nash solution was proposed by Nash (1950), who also established the notable fact that it is the only solution that satisfies all the conditions so far.

Proposition 22.E.1: The Nash solution is the only bargaining solution that is independent of utility origins and units, Paretian, symmetric, and independent of irrelevant alternatives.²⁸

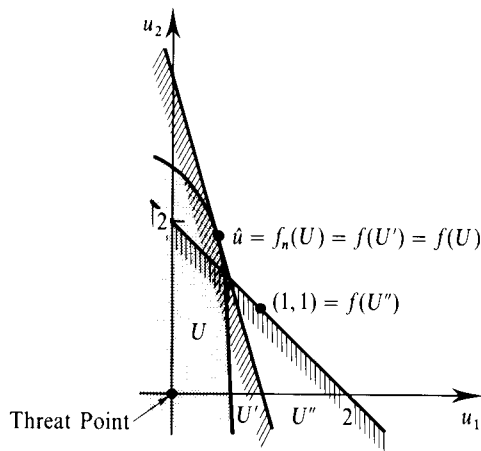
Proof: We have already shown in the discussion of Example 22.E.4 that the Nash solution satisfies the properties claimed.

To establish the converse, suppose we have a candidate solution $f(\cdot)$ satisfying all the properties. By the independence of utility origins, we can assume, as we have done so far, that $f(\cdot)$ is defined on sets where the threat point has been normalized to the origin. Given now an arbitrary U , let $\hat{u} = f_n(U)$ and consider the sets

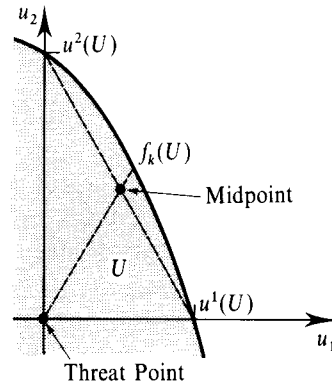
$$U' = \{u \in \mathbb{R}^I : \sum_i u_i / \hat{u}_i \leq I\} \quad \text{and} \quad U'' = \{u \in \mathbb{R}^I : \sum_i u_i \leq I\}.$$

27. To repeat in more geometric terms: the hyperplane with normal $(\eta_1, \dots, \eta_I)$ passing through $\hat{u}$ leaves U below it (because of the utilitarian property). Thus, it suffices to show that the set $\{u : \sum_i \ln u_i \leq \sum_i \ln \hat{u}_i\}$ lies above the hyperplane. But note that this follows from the fact that, because of the egalitarian property, $(\eta_1, \dots, \eta_I)$ is proportional to $(1/\hat{u}_1, \dots, 1/\hat{u}_I)$, which is the gradient of the concave function $\sum_i \ln u_i$ at $\hat{u}$.

28. Note that we do not assume individual rationality explicitly: it turns out to be implied by the other conditions.

**Figure 22.E.7 (left)**

The Nash solution is determined uniquely from the independence of utility origins and units, symmetry, Pareto, and independence of irrelevant alternatives properties (Proposition 22.E.1).

**Figure 22.E.8 (right)**

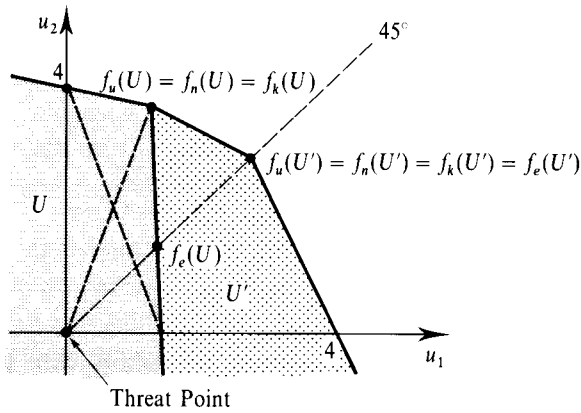
The Kalai-Smorodinsky solution for bargaining problems.

See Figure 22.E.7 for an illustration in the case $I = 2$. Note that $U \subset U'$ because the concave function $\sum_i \ln u_i$ has gradient $(1/\hat{u}_1, \dots, 1/\hat{u}_I)$ at $\hat{u}$, the point where it reaches its maximum value in the convex set U . The set U'' is symmetric and, therefore, by the symmetry and Paretian properties, we conclude that $f(U'') = (1, \dots, 1)$. By the independence of utility units, it then follows that $f(U') = (\hat{u}_1, \dots, \hat{u}_I) = \hat{u}$ [observe that $u \in U''$ if and only if $(\hat{u}_1 u_1, \dots, \hat{u}_I u_I) \in U'$]. Finally, since $\hat{u} \in U$ and $U \subset U'$, the independence of irrelevant alternatives property yields $f(U) = \hat{u} = f_n(U)$, which is the result we wanted. ■

Example 22.E.4: Kalai-Smorodinsky Solution. This will be an example of a solution that does not satisfy the independence of irrelevant alternatives property. It was proposed by Kalai and Smorodinsky (1975). Given a bargaining problem $U \subset \mathbb{R}^I$, denote by $u^i(U) \in \mathbb{R}$ the maximum utility value that agent i could attain by means of some vector in $U \cap \mathbb{R}_+^I$. See Figure 22.E.8 for the case $I = 2$. To motivate the solution suppose that agent i has all the bargaining power (i.e., she can make a take-or-leave offer to the remaining agents). Then the outcome would give $u^i(U)$ to agent i and nothing to the remaining ones.²⁹ Hence, we could regard the numbers $u^i(U)$ as rough measures of the contributions of the respective agents to the cooperative endeavor and argue, perhaps, that if cooperation takes place then the solution should be the Pareto optimal allocation where the utilities of the different agents are proportional to $(u^1(U), \dots, u^I(U))$; in other words, where utilities are proportional to the expected utilities that would obtain if we chose with equal probability the agent making a take-or-leave offer. This is precisely the Kalai-Smorodinsky solution $f_k(U)$. Its construction is indicated in Figure 22.E.8.

The Kalai-Smorodinsky solution satisfies the Paretian and the symmetry properties. As with the Nash solution, it *does not involve interpersonal comparisons of utilities*. However, it is different from the Nash solution and, therefore (by Proposition 22.E.1), it cannot satisfy the IIA property. In Exercise 22.E.4 you are asked to verify all this. ■

29. We neglect cases where the utility vector that gives $u^i(U)$ to agent i and nothing to the remaining ones is Pareto dominated.

**Figure 22.E.9**

Lack of monotonicity of the utilitarian, Nash, and Kalai-Smorodinsky bargaining solutions.

The conditions summarized so far are by no means an exhaustive list of the properties that have been found to be of interest in the study of the bargaining problem. We conclude with a few informal comments about some others.

(i) *Linearity, or decomposability, properties.* Suppose that given two bargaining problems, $U \subset \mathbb{R}^I$ and $U' \subset \mathbb{R}^I$, we consider $\alpha U + (1 - \alpha)U' \subset \mathbb{R}^I$, for $\alpha \in [0, 1]$; we may think of this as, for example, a randomization between the two problems. Then we may want $f(\alpha U + (1 - \alpha)U') = \alpha f(U) + (1 - \alpha)f(U')$; that is, we may wish that all agents be indifferent between coming to a settlement before or after the resolution of uncertainty. This is a strong requirement, and none of the solutions we have studied satisfies it. In fact, it can be shown that, essentially (i.e., with a few other weak conditions), it is only satisfied by the modified version of the utilitarian solution that does not impose individual rationality. The same conclusion is reached if we consider $U + U'$ and ask that the overall settlement $f(U + U')$ equals the sequential settlement $f(U' + \{f(U)\})$, $f(U)$. Recall that by the IUO condition the last expression equals $f(U') + f(U)$.

(ii) *Monotonicity properties.* A bargaining solution $f(\cdot)$ is *monotone* if $f(U) \leq f(U')$ whenever $U \subset U'$, that is, if whenever the utility possibility set expands (keeping the threat point fixed at the origin), it is to everyone's advantage. The monotonicity requirement is stronger than may appear at first sight because the utility possibility set may expand in ways that are very asymmetric across agents. In fact, you can verify in Figure 22.E.9 that neither the utilitarian, nor the Nash, nor the Kalai-Smorodinsky solutions satisfy monotonicity. On the other hand, the egalitarian solution clearly does. In Exercise 22.E.5 you are invited to verify that the egalitarian solution is essentially the only symmetric and Paretian bargaining solution that satisfies the monotonicity condition. In Exercise 22.E.6 you can also check that the Kalai-Smorodinsky solution for $I = 2$ is characterized, with the IUO, IUU, P, and S properties, by a certain condition of partial monotonicity.

(iii) *Consistency properties.* This type of property concerns the mutual fit of the bargaining solutions when we apply them to problems with different numbers of agents. Let $f^I(\cdot)$ be a family of bargaining solutions (one for every set of agents I). Suppose, to be specific, that we start with $I = \{1, 2, 3\}$. Take any i , say $i = 1$, and imagine that, conditional on final cooperation, we give agent 1 the utility level $f_1^I(U)$, but that after making this commitment we reopen the negotiation between the two remaining agents. These two agents then have to find a settlement between themselves in the set $U' = \{(u_2, u_3) : (f_1^I(U), u_2, u_3) \in U\} \subset \mathbb{R}^2$. It is then natural to apply the solution in our family, that is, $f^{I \setminus \{1\}}(U')$. Our family is consistent if

$f^{\wedge(1)}(U') = (f_2^I(U), f_3^I(U))$. In words: the renegotiation leads exactly to the outcome of the initial negotiation. The utilitarian and the Nash solution are consistent (in general, any solution obtained by maximizing a generalized utilitarian SWF is consistent; see Exercise 22.E.7). The Kalai–Smorodinski solution is not (verify this in Exercise 22.E.8). It is an interesting, and nontrivial, fact that the consistency axiom can be used instead of IIA in the characterization of the Nash solution [see Lensberg (1987) and Thomson and Lensberg (1992)].

22.F Coalitional Bargaining: The Shapley Value

The analysis of Section 22.E is limited in one important respect: it does not contemplate the possibility of situations intermediate between the full cooperation of all agents and the complete breakdown represented by the threat-point outcome. When there are more than two agents this is definitely restrictive. In this section, we make allowance for the possibility of *partial cooperation* and discuss how this may influence the eventual distribution of the gains from cooperation.

We are given a set I of agents and we proceed by specifying a possible set of utility outcomes to every potential subset of cooperators $S \subset I$. When $S \neq I$ these utility outcomes are interpreted as a description of what may occur if bargaining breaks down and the members of S end up cooperating among themselves. For the purpose of simplicity, we limit ourselves to the case where utility is comparable across agents (we fix individual units to have the same social utility value) and freely transferable among them. We can then represent the total amount of utility available to the members of $S \subset I$, if they cooperate, by a number $v(S)$ or, equivalently, by the utility possibility set $\{u \in \mathbb{R}^S: \sum_{i \in S} u_i \leq v(S)\}$. In cooperative game theory, which we reviewed in Appendix A of Chapter 18, the rule that assigns $v(S)$ to every $S \subset I$ is known as a *game in characteristic form*, a subset $S \subset I$ is usually referred to as a *coalition*, and the number $v(S)$ is known as the *worth* of the coalition S .

The situation in which there are no gains from partial cooperation is captured by a characteristic form where $v(S) = \sum_{i \in S} v(i)$ for every $S \neq I$. When we interpret the vector $(v(1), \dots, v(I))$ as the threat point, this situation reduces to the bargaining problem studied in Section 22.E. In the current world of transferable utility, the egalitarian, Nash, and Kalai–Smorodinsky bargaining solutions³⁰ lead to the same proposal: The gains from cooperation should be split equally among the agents; that is, agent $i \in I$ should get

$$v(i) + \frac{1}{I} \left(v(I) - \sum_{h \in I} v(h) \right).$$

In fact, any bargaining solution that is independent of utility origins, Paretian, and symmetric makes this recommendation (Exercise 22.F.1).

The question we will try to answer in this section is the following: *Assume that, in an environment of games in characteristic form, all the members of I decide on cooperation, and therefore on distributing $v(I)$ among themselves. What is then the proper generalization of the equal split solution?* It stands to reason that the solution will have to reflect, in some manner, the numbers $v(S)$, $S \subset I$, since these incorporate

30. The utilitarian solution is not uniquely defined in the transferable utility case.

the information on how valuable the contribution of a particular agent is compared to that of another.

Definition 22.F.1: Given the set of agents I , a *cooperative solution* $f(\cdot)$ is a rule that assigns to every game $v(\cdot)$ in characteristic form a utility allocation $f(v) \in \mathbb{R}^I$ that is feasible for the entire group, that is, such that $\sum_i f_i(v) \leq v(I)$.

Reiterating the analytical strategy of Section 22.E, we continue by stating a number of desirable properties for a solution. The first three are merely variations of properties already encountered above.

Definition 22.F.2: The cooperative solution $f(\cdot)$ is *independent of utility origins and of common changes of utility units* if, whenever we have two characteristic forms $v(\cdot)$ and $v'(\cdot)$ such that $v(S) = \beta v'(S) + \sum_{i \in S} \alpha_i$ for every $S \subset I$ and some numbers $\alpha_1, \dots, \alpha_I$, and $\beta > 0$, it follows that $f(v) = \beta f(v') + (\alpha_1, \dots, \alpha_I)$.

From now on we assume the property of Definition 22.F.2. Because of it we can, in particular, normalize $v(\cdot)$ to $v(i) = 0$ for all i .

Definition 22.F.3: The cooperative solution $f(\cdot)$ is *Paretian* if $\sum_i f_i(v) = v(I)$, for every characteristic form $v(\cdot)$.

Definition 22.F.4: The cooperative solution $f(\cdot)$ is *symmetric* if the following property holds: Suppose that two characteristic forms, $v(\cdot)$ and $v'(\cdot)$ differ only by a permutation $\pi: I \rightarrow I$ of the names of the agents; that is, $v'(S) = v(\pi(S))$ for all $S \subset I$. Then the solution also differs only by this permutation; that is, $f_i(v') = f_{\pi(i)}(v)$ for all i .

The property defined next, in Definition 22.F.5, underlines the fact that we are trying to solve not the welfare-theoretic problem of distributing total utility equitably but rather the more limited problem of distributing equitably the *surplus* that can be attributed to the cooperation among agents, given the realities of the particular bargaining situation captured by the characteristic form.

Definition 22.F.5: The cooperative solution $f(\cdot)$ satisfies the *dummy axiom* if, for all games $v(\cdot)$ and all agents i such that $v(S \cup \{i\}) = v(S)$ for all $S \subset I$, we have $f_i(v) = v(i) (= 0)$. In words: If agent i is a *dummy* (i.e., does not contribute anything to any coalition), then agent i does not receive any share of the surplus.

There are a number of cooperative solutions that satisfy the above properties. The most important is the *Shapley value* [proposed by Shapley (1953)]. We refer to Appendix A of Chapter 18 on cooperative game theory for examples, motivation, and extended discussion of this concept. Here we limit ourselves to offering a definition.

Suppose that we consider an arbitrary ordering of the agents or, formally, an arbitrary permutation π of the names $\{1, \dots, I\}$. Then

$$g_{v,\pi}(i) = v(\{h: \pi(h) \leq \pi(i)\}) - v(\{h: \pi(h) < \pi(i)\})$$

represents how much agent i contributes when she joins the group of her predecessors in the ordering. This is the amount the predecessors would agree to pay i if she had

all the negotiating power, that is, if she could make a take-it-or-leave-it offer.³¹ Note that $\sum_i g_{v,\pi}(i) = v(I)$ for all permutations π .

The agents do not come to us ordered. They all stand on the same footing. We may account for this by giving every agent the same chance of being in any position, thereby making all positions equally likely. Equivalently, we could take the (equal weighting) average of agent i contributions over all permutations π (there are $I!$ of these). This is precisely the Shapley value solution.

Definition 22.F.6: The Shapley value solution $f_s(\cdot)$ is defined by

$$f_{si}(v) = \frac{1}{I!} \sum_{\pi} g_{v,\pi}(i) \quad \text{for every } i. \quad (22.F.1)$$

It is simple to verify that $f_s(\cdot)$ satisfies all the properties listed so far (see Exercise 22.F.2). For further discussion, see Exercises 22.F.3 and 22.F.4 (and, to repeat, Appendix A of Chapter 18). In Exercise 22.F.5 we describe another cooperative solution (the nucleolus).

The Cost-Allocation Problem

The following discussion is in the nature of an appendix as there is no immediate conceptual connection with the previous material. The link is that we again make use of the Shapley value.³²

Suppose that we have a set I of *projects* and that a policy decision to carry them forward has already been taken. For some reason (e.g., accounting or financing rules), the total cost $C(I)$ must be allocated exactly to the different projects; that is, we must specify $(c_1, \dots, c_I)$ such that $\sum_i c_i = C(I)$. What is a reasonable way to determine such *cost allocation rules*? This is the cost-allocation problem.

We must first of all emphasize that, from the point of view of first-best optimality, the cost allocation rules *should not* be used to guide investment, that is, to decide which projects to carry out. Loosely speaking, we have seen in Section 16.G that the correct efficiency prices for inputs (note that we can think of projects as inputs to the production function for welfare) do not need to cover total cost exactly (see Exercises 22.F.6 and 22.F.7). Because we wish to avoid the temptation to use cost allocation rules in this way we insist that the set of projects to be implemented be exogenously given.

An alternative would be to recognize that the cost-covering constraint confers to the welfare problem a second-best nature. That is, we could attempt to maximize social welfare subject to the condition that input (i.e., project) prices must be fixed at levels that exactly cover costs. This approach was taken by Boiteux (1956) in the context of the theory of the regulated firm, and the solution is closely related to Ramsey pricing (see Example 22.B.2 and Exercise 22.F.6).

Any welfare-theoretic approach, however, would need to use information on individual preferences. If this cannot, or should not, be done, we are still left with an unresolved the problem. A suggested approach proceeds then as follows: suppose that we have information on the cost of every subset of projects (this is far from an innocuous demand); that is, we know $C(T)$ for every $T \subset I$. Then, formally, $C(\cdot)$ is a cooperative game in characteristic form

31. In other words, an offer that, if rejected by some predecessor, would permanently preclude the possibility of agent i or any successor from joining the coalition of the predecessors. To verify the informal claim that the offer will be $g_{v,\pi}(i)$, determine how much the last agent in the ordering will get and proceed by backward induction.

32. See Young (1994) for an introductory account to cost-allocation and related problems.

and we could resort to the Shapley value as a way to split $C(I)$ among the different projects. An example may help to clarify the point.

Example 22.F.1: This is a favorite example for academics. A professor of economics based in the United States is planning a grand tour of Europe that will take her to three universities, one in Britain (B), one in Spain (S), and one in Germany (G). The total air fare comes to 1600 dollars. How is this to be reimbursed by the three institutions? Suppose that after some research it turns out that $C(B) = C(S) = C(G) = 800$, $C(BS) = C(BG) = 1000$, and $C(SG) = 1400$. The Shapley value calculation (carry it out!) then gives $c_B = 400$ and $c_S = c_G = 600$. This split does indeed seem to reflect well the comparative ease of managing a side trip to Britain if already going to some of the other destinations. ■

REFERENCES

- d'Aspremont, C., and L. Gevers. (1977). Equity and the informational basis of collective choice. *Review of Economic Studies* 44: 199–209.
- Atkinson, A. (1973). How progressive should income-tax be? In *Economic Justice, Selected Readings*, edited by E. Phelps. London: Penguin Books.
- Atkinson, A., and J. Stiglitz. (1980). *Lectures on Public Economics*. New York: McGraw-Hill.
- Bergson, A. (1938). A reformulation of certain aspects of welfare economics. *Quarterly Journal of Economics* 52: 310–34.
- Boiteux, M. (1956). Sur la gestion des monopoles publics astreints à l'équilibre budgétaire. *Econometrica* 24: 22–40. [Translated in *Journal of Economic Theory* (1991) 3: 219–40.]
- Harsanyi, J. (1955). Cardinal welfare, individual ethics, and interpersonal comparability of utility. *Journal of Political Economy* 61: 309–21. [Also in Phelps (1973).]
- Guesnerie, R. (1995). *A Contribution to the Pure Theory of Taxation*, Cambridge, U.K.: Cambridge University Press.
- Kalai, E., and M. Smorodinsky. (1975). Other solutions to Nash's bargaining problem. *Econometrica* 43: 513–18.
- Laffont, J.-J. (1988). *Fundamentals of Public Economics*. Cambridge, Mass.: MIT Press.
- Lipsey, R. C., and K. Lancaster. (1956). The general theory of the second best. *Review of Economic Studies* 24: 11–32.
- Lensberg, T. (1987). Stability and collective rationality. *Econometrica* 55: 935–62.
- Maskin, E. (1978). A theorem on utilitarianism. *Review of Economic Studies* 42: 93–96.
- Moulin, H. (1988). *Axioms of Cooperative Decision Making*. Cambridge, U.K.: Cambridge University Press.
- Nash, J. F. (1950). The bargaining problem. *Econometrica* 28: 155–62.
- Phelps, E., ed. (1973). *Economic Justice, Selected Reading*. London: Penguin Books.
- Ramsey, F. (1927). A contribution to the theory of taxation. *Economic Journal* 37: 47–61.
- Rawls, J. (1971). *A Theory of Justice*. Cambridge, Mass.: Harvard University Press.
- Roberts, K. (1980). Possibility theorems with interpersonally comparable welfare levels. *Review of Economic Studies* 47: 409–20.
- Roth, A. (1979). *Axiomatic Models of Bargaining*. New York: Springer-Verlag.
- Samuelson, P. (1947). *Foundations of Economic Analysis*. Cambridge, Mass.: Harvard University Press.
- Sen, A. (1977). On weights and measures: informational constraints in social welfare analysis. *Econometrica* 45: 1539–72.
- Shapley, L. (1953). A value for n -person games. In *Contributions to the Theory of Games II. Annals of Mathematics Studies*, 28, edited by H. Kuhn, and A. Tucker. Princeton, N.J.: Princeton University Press.
- Starrett, D. A. (1988). *Foundations of Public Economics*. Cambridge, U.K.: Cambridge University Press.
- Thomson, W., and T. Lensberg. (1992). *The Theory of Bargaining with a Variable Number of Agents*. Cambridge, U.K.: Cambridge University Press.

- Thomson, W. (1995). Cooperative models of bargaining. In *Handbook of Game Theory*, Vol. II, edited by R. Aumann, and S. Hart. Amsterdam: North-Holland.
- Young, H. P. (1994). *Equity. In Theory and Practice*. Princeton, N.J.: Princeton University Press.

EXERCISES

22.B.1^A Give sufficient conditions for the convexity of the first-best utility possibility set in the context of the exchange economies of Example 22.B.1.

22.B.2^A Derive the first-order conditions stated in Example 22.B.2.

22.B.3^A Derive the first-order conditions (22.B.2) of Example 22.B.3.

22.B.4^B Show as explicitly as you can that the utility possibility set of Example 22.B.4 may not be convex.

22.C.1^A Suppose that the utility possibility set $U \subset \mathbb{R}^I$ is symmetric and convex. Show that the social optimum of an increasing, symmetric, strictly concave social welfare function $W(\cdot)$ assigns the same utility values to every agent. [Note: A set U is symmetric if $u \in U$ implies $u' \in U$ for any u' obtained from u by a permutation of its entries.] Observe that the same conclusion obtains if $W(\cdot)$ is allowed to be just concave, as in the utilitarian case, but U is required to be strictly convex.

22.C.2^A Suppose that we contemplate a decision maker in an original position (or *ex-ante*, or *behind the veil of ignorance*) before the occurrence of a state of the world that will determine which of I possible identities the decision maker will have. There is a finite set X_i of possible final outcomes in identity i . Denote $X = X_1 \times \cdots \times X_I$.

(a) Appeal to the theory of state-dependent utility presented in Section 6.E to justify a utility function on X of the form

$$U(x_1, \dots, x_I) = u_1(x_1) + \cdots + u_I(x_I).$$

Interpret and discuss the implications of this utility function for the usage of a purely utilitarian social welfare function.

(b) Suppose that $X_1 = \cdots = X_I$ and the preference relation on X defined by the utility function in (a) is symmetric. What does this imply for the form of the utility function? Discuss and interpret.

22.C.3^B We have N final social outcomes and we consider a set of alternatives X that is the set of lotteries over these outcomes. An alternative can be represented by the list of probabilities assigned to the different final outcomes, that is, $p = (p_1, \dots, p_N)$ where $p_n \geq 0$ for every n and $p_1 + \cdots + p_N = 1$.

We assume that we are given a social preference relation $\succsim$ on X that is continuous and conforms to the independence axiom. Thus, it can be represented by a utility function of the expected utility form

$$U(p) = u_1 p_1 + \cdots + u_N p_N.$$

From now on we assume that this social utility function $U(\cdot)$ defined on X is given.

(a) Suppose that there are two final outcomes and that they are specified by which of two individuals will receive a certain indivisible object. Suppose also that social preferences are symmetric in the sense that there is social indifference between the lottery that gives the object to individual 1 for sure and the lottery that gives the object to individual 2 for sure. Show that all the lotteries must then be socially indifferent. Discuss and interpret this conclusion.

Is it plausible to you? If you wanted to escape from it, how would you do it? What does this all say about the independence axiom as applied to social decisions?

Suppose now that there are I agents and that in addition to the social utility function $U(\cdot)$ we are also given I individual preference relations $\succsim_i$ defined on the same set of lotteries X . We assume that they are also represented by utility functions of the expected utility form

$$U_i(p) = u_{i1}p_1 + \cdots + u_{iN}p_N \quad \text{for } i = 1, \dots, I.$$

We say that the social utility function $U(\cdot)$ is *Paretian* if we have $U(p) > U(p')$ whenever $U_i(p) > U_i(p')$ for every i .

(b) Consider a case with $N = 3$ and $I = 2$ and illustrate, in the 2-dimensional simplex of lotteries, how the indifference map of the utility functions of the two agents and of the social utility function fit together when the social utility function is Paretian.

(c) Exhibit a case where the Paretian condition determines uniquely the social indifference map (recall that we are always assuming the independence axiom for social preferences!). Argue, however, that in general the Paretian condition does not determine uniquely the social indifference map. In fact, exhibit an example where any social utility function is Paretian.

(d) Argue (you can restrict yourself to $N = 3$ and $I = 2$) that if the social utility function $U(p)$ is Paretian then it can be written in the form

$$U(p) = \beta_1 U_1(p) + \cdots + \beta_I U_I(p)$$

where $\beta_i \geq 0$ for every i and $\beta_i \neq 0$ for some i . What does this conclusion say for the usage of a purely utilitarian social welfare function? Interpret the β_i weights, as well as the fact that they need not be equal across individuals.

22.C.4^A The *leximin* ordering, or preference relation, on $\mathbb{R}^I$ has been mentioned in footnote 11 of this chapter when discussing the Rawlsian SWF. It is formally defined as follows. Given a vector $u = (u_1, \dots, u_I)$ let $u^r \in \mathbb{R}^I$ be the vector that is the *nondecreasing rearrangement* of u . That is, the entries of u^r are in nondecreasing order and its numerical values (multiplicities included) are the same as for u . We then say that the vector u is at least as good as the vector $\hat{u}$ in the leximin order if u^r is at least as good as $\hat{u}^r$ in the lexicographic ordering introduced in Example 3.C.1.

(a) Interpret the definition of the leximin as a refinement of the Rawlsian preference relation.

(b) Show that the leximin ordering cannot be represented by a utility function. It is enough to show this for $I = 2$.

(c) (Harder) Show that the social optimum of a leximin ordering is a Pareto optimum. You can limit yourself to the case $I = 3$.

22.C.5^B Consider the constant elasticity family of social welfare functions (Example 22.C.4). Argue that $W_\rho(u) \rightarrow \text{Min}\{u_1, \dots, u_I\}$ as $\rho \rightarrow \infty$.

22.C.6^A Suppose that U and U' are utility possibility sets and that we associate with them Pareto optimal utility outcomes $\bar{u} \in U$ and $\bar{u}' \in U'$, respectively. Show graphically that:

(a) It is possible for U' to pass the strong compensation test over U and yet for the outcome with U' to be worse than the outcome with U , as measured by the purely utilitarian SWF.

(b) If the utility possibility sets are derived from a quasilinear economy and U' passes the weak compensation test over U , then it also passes the strong compensation test and, moreover, the outcome for U' is a utilitarian improvement over the outcome for U . Is this conclusion valid if we evaluate social welfare by a nonutilitarian SWF?

22.C.7^B Construct an explicit example of two Edgeworth box economies, differing only in their distributions of the initial endowments, such that the utility possibility set of each one

passes the weak compensation test over the utility possibility set of the other, when the utility outcome in the latter is chosen to correspond to one of its competitive equilibria.

22.C.8^A Suppose we have two utility possibility sets U, U' with respective outcomes $u \in U$ and $u' \in U'$. We say that (U', u') passes the *Kaldor compensation test* over (U, u) if U' passes the weak compensation test over (U, u) and U does not pass the weak compensation test over (U', u') .

(a) For $I = 2$, represent graphically a situation where Kaldor comparability is possible and one where it is not.

(b) Observe that Kaldor comparability is asymmetric. Define your terms.

(c) Show that Kaldor comparability may not be transitive.

22.D.1^B In this exercise we verify the indispensability of the assumptions of Proposition 22.D.1.

(a) Suppose there are three agents and only two alternatives, that is, $X = \{x, y\}$. The social welfare functional is given by

$$\begin{aligned} x F(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3) y & \quad \text{if and only if} \quad \tilde{u}_i(x) \geq \tilde{u}_i(y) \text{ for every } i \\ \text{and} \\ y F(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3) x & \quad \text{if and only if} \quad \tilde{u}_i(y) \geq \tilde{u}_i(x) \text{ for at least one } i. \end{aligned}$$

Check that the social preference relation is always complete, that the social welfare functional cannot be represented by means of a social welfare function, and that only the condition on the number of alternatives fails from Proposition 22.D.1.

(b) Now we have three agents and three alternatives, that is, $X = \{x, y, z\}$. The social welfare functional is given by

$$x F_p(\tilde{u}_1, \dots, \tilde{u}_I) y F_p(\tilde{u}_1, \dots, \tilde{u}_I) z$$

for every $(\tilde{u}_1, \dots, \tilde{u}_I) \in \mathcal{U}^I$. Show that, again, no representation by means of a social welfare function is possible and that, of the assumptions of Proposition 22.D.1, only the Paretian property fails to be satisfied.

(c) Exhibit an example in which the only condition of Proposition 22.D.1 that fails to be satisfied is pairwise independence.

22.D.2^A Carry out the verification requested in the second paragraph of the proof of Proposition 22.D.1.

22.D.3^A In text.

22.D.4^A A social welfare functional F is *lexically dictatorial* if there is a list of $n > 0$ agents $h_1, \dots, h_n$ such that the strict preference of h_1 prevails socially, the strict preference of h_2 prevails among the alternatives for which h_1 is indifferent, and so on.

(a) Show that if F is lexically dictatorial then F is Paretian, is pairwise independent, and does not allow for interpersonal comparisons of utility.

(b) Under what conditions can a social welfare functional that is lexically dictatorial be generated from a social welfare function?

(c) Show that if a dictatorial social welfare functional is generated from a social welfare function $W(u) = \sum_i b_i u_i$, then $b_i = 0$ for every i distinct from the dictator.

22.D.5^C Complete the proof of Arrow's impossibility theorem along the lines suggested in the last paragraph prior to the small-type text at the end of Section 22.D. (Assume that Proposition

22.D.3 is valid under the weakened assumption that F is generated from a social preference relation on $\mathbb{R}^I$.)

22.D.6^B This exercise is concerned with social welfare functions satisfying expression (22.D.1).

(a) Show that the nonsymmetric utilitarian function $W(u) = \sum_i b_i u_i$ can be written in the form (22.D.1).

(b) Show that if $W(\cdot)$ is symmetric and $g(0) = 0$ then $g(\cdot) \geq 0$.

(c) Show that the symmetric Rawlsian social welfare function $W(u) = \text{Min} \{u_1, \dots, u_I\}$ can be written in the form (22.D.1). What about nonsymmetric Rawlsian social welfare functions? [Hint: Check the condition of invariance to common changes of origins.]

(d) Give other examples satisfying (22.D.1), in particular, examples with $g(\cdot) \geq 0$ and intermediate between the utilitarian and the Rawlsian cases. Interpret them.

(e) Argue that if in (22.D.1) the function $g(\cdot)$ is homogeneous of degree one and differentiable, then it must be linear (and so we are back to the utilitarian case).

22.D.7^B Consider the constant elasticity family of social welfare functions studied in Example 22.C.4.

(a) Show that the social welfare functionals derived from SWFs in this family are invariant to common changes of units.

(b) Show that the only members of this family which are also invariant to common changes of origins, and therefore admit a representation in the form (22.D.1), are the purely utilitarian (i.e., $\rho = 0$) and the Rawlsian (i.e., $\rho = \infty$).

22.D.8^B This is an exercise on the property of invariance to common ordinal transformation.

(a) Show that the symmetric, Rawlsian social welfare function satisfies the property.

(b) Show that the anti-Rawlsian function $W(u) = \text{Max} \{u_1, \dots, u_I\}$ also satisfies it.

(c) Show that the property is satisfied for dictatorial social welfare functionals.

(d) (Harder) Suppose that $I = 2$ and $W(u) = W(u')$ for two vectors $u, u' \in \mathbb{R}^2$, with $u'_1 < u_1 < u_2 < u'_2$. Assume also that $W(\cdot)$ is increasing. Show that the induced social welfare functional cannot be invariant to identical ordinal transformations. From this, argue informally (you can do it graphically) that for $I = 2$ a continuous, increasing social welfare function that is also invariant to identical ordinal transformations must be either dictatorial, Rawlsian, or anti-Rawlsian.

22.E.1^A Verify that the bargaining solutions in Examples 22.E.1 to 22.E.4 are independent of utility origins, Paretian, symmetric, and individually rational. It is enough if you do so for $I = 2$.

22.E.2^A State nonsymmetric versions of the four bargaining solutions studied in Section 22.E (egalitarian, utilitarian, Nash, and Kalai–Smorodinsky). Motivate them.

22.E.3^B This is an exercise on the Nash solution.

(a) Verify that for $I = 2$, $f_n(U)$ is the boundary point of U through which we can draw a tangent line with the property that its midpoint in the positive orthant is precisely the given boundary point $f_n(U)$.

(b) Verify that if $U \subset \mathbb{R}^I$ is a bargaining problem then there are rescaling units for the individual utilities with the property that the Nash solution becomes simultaneously egalitarian and utilitarian.

22.E.4^A Verify that the Kalai–Smorodinsky solution satisfies the property of independence of utility units but violates the property of independence of irrelevant alternatives. You can restrict yourself to $I = 2$.

22.E.5^B This is an exercise on the monotonicity property.

(a) Show that the egalitarian solution is the only bargaining solution that is independent of utility origins, Paretian, symmetric and monotonic. [Hint: Consider first a family of symmetric utility possibility sets with linear boundaries. Notice then that for any two sets U, U' we always have $U \cap U' \subset U$ and $U \cap U' \subset U'$.]

(b) (Harder) Suppose that $f(\cdot)$ is a bargaining solution that is independent of utility origins, Paretian, and strongly monotonic [if $U \subset U'$ then $f(U) \leq f(U')$ and, in addition, if $f(U)$ is interior to U' then $f(U) \ll f(U')$]. Show that there is a curve in $\mathbb{R}^I$ starting at the origin and strictly increasing such that, for every U , $f(U)$ is the intersection point of the boundary of U with this curve. You can restrict yourself to the case $I = 2$.

22.E.6^C Let $I = 2$. A bargaining solution $f(\cdot)$ is *partially monotone* if when $U \subset U'$ and $u^i(U) = u^i(U')$, that is, U' expands U only in the direction of agent $j \neq i$, we have $f_j(U') \geq f_j(U)$ for $j \neq i$. Argue that the Kalai–Smorodinsky solution is characterized by the following properties: independence of utility origins and units, Pareto, symmetry, and partial monotonicity. [Hint: use sets U such that $U' \subset U$ and $u^1(U) = u^1(U')$, $u^2(U) = u^2(U')$.]

22.E.7^A Consider a family of bargaining solutions $f^I(\cdot)$ such that, for every set of agents I , $f^I(\cdot)$ is independent of utility origins and is generated by maximizing the social welfare function $\sum_i g(u_i)$ on normalized bargaining problems $U \subset \mathbb{R}^I$, where $g(\cdot)$ is increasing, strictly concave, and independent of the particular I considered. Show that the family f^I is consistent.

22.E.8^C Show by example that the Kalai–Smorodinsky solution is not consistent. It is enough to consider three agents and its subgroups of two agents.

22.E.9^A This exercise is aimed at showing the independence of the assumptions of Proposition 22.E.1. To this effect, give five examples such that for each of the five assumptions of Proposition 22.E.1 there is one of the examples that violates this assumption but satisfies the remaining four.

22.E.10^A Give an example of a utilitarian bargaining solution (Example 22.E.2) that violates the property of independence of irrelevant alternatives. [Hint: It suffices to consider $I = 2$. Also, the violation should involve sets U that are convex but not strictly convex.]

22.E.11^C Go back to the infinite horizon Rubinstein's bargaining model discussed in the Appendix A to Chapter 9 (specifically, Example 9.AA.2). The only modification is that the two agents are risk averse on the amount of money they get. That is, each has an increasing, concave, differentiable utility function $u_i(m_i)$ on the nonnegative amounts of money that they receive. The factor of discount $\delta < 1$ is the same for the two agents. Also $u_i(0) = 0$. The total amount of money is m .

(a) Write down the equations for a subgame perfect Nash equilibrium (SPNE) in stationary strategies. Argue that there is a single configuration of utility payoffs that can be obtained as payoffs of a SPNE in stationary strategies.

(b) Consider the utility possibility set

$$U = \{(u_1(m_1), u_2(m_2)) \in \mathbb{R}^2 : m_1 + m_2 = m\} - \mathbb{R}_+^2.$$

Show that if δ is close to 1 then the payoffs of a SPNE in stationary strategies are nearly equal to the Nash bargaining solution payoffs.

(c) (Harder) Argue that every payoff configuration of a SPNE can be obtained as the payoff configuration of a SPNE in stationary strategies. Thus, the uniqueness result presented in Example 9.AA.2 extends to the case in which the agents have strictly concave, possibly different, utility functions for money.

22.F.1^A Show that in the transferable utility case any bargaining solution that is invariant to independent changes of origin, symmetric, and Paretian divides the gains from cooperation equally among the agents.

22.F.2^A Show that the Shapley value cooperative solution presented in Section 22.F satisfies the following properties: invariance to independent changes of utility origins, invariance to common changes of utility units, Paretian, symmetry, and the dummy axiom.

22.F.3^A Suppose that for a given set of agents I we take two characteristic forms v and v' and consider their sum $v + v'$; that is, $v + v'$ is the characteristic form where $(v + v')(S) = v(S) + v'(S)$ for every $S \subset I$.

(a) Verify that the Shapley value is *linear* in the characteristic form; that is, $f_{si}(v + v') = f_{si}(v) + f_{si}(v')$ for all v, v' and i .

(b) Interpret the linearity property as a postulate that agents are indifferent to the timing of resolution of uncertainty when we randomize among bargaining situations.

22.F.4^C The linearity property of the previous exercise can be restated in a perhaps more intuitive form. We say that a characteristic form $v(\cdot)$ is a *unanimity game* if for some $T \subset I$ we have that $v(S) = v(T)$ if $T \subset S$, and $v(S) = 0$ otherwise (thus, the bargaining situations of Section 22.E correspond to $T = I$).

(a) Show that the independence of utility origins and invariance to common changes of utility units, Pareto, symmetry, and dummy axiom properties imply that, for a unanimity game $v(\cdot)$, any cooperative solution $f(\cdot)$ assigns the values $f_i(v) = (1/T)v(I)$ if $i \in T$, and $f_i(v) = 0$ otherwise.

(b) We say that the cooperative solution $f(\cdot)$ is *weakly linear* if for any v and v' differing only by a unanimity game [i.e., there is $T \subset I$ and $\alpha \in \mathbb{R}$ such that $v'(S) = v(S) + \alpha$ if $T \subset S$, and $v'(S) = v(S)$ otherwise] we have that $f_i(v') = f_i(v) + \alpha/T$ if $i \in T$, and $f_i(v') = f_i(v)$ otherwise. Show that if, in addition to the properties listed in (a), the cooperative solution $f(\cdot)$ is weakly linear, then it is fully linear, that is, $f(v + v') = f(v) + f(v')$ for any two characteristic forms v and v' .

(c) Show that the Shapley value is the only cooperative solution that satisfies the following properties: independence of utility origins and invariance to common changes of utility units, Paretian, symmetry, dummy axiom, and linearity.

22.F.5^C In this exercise we describe another cooperative solution for a game in characteristic form: the *nucleolus*. For simplicity we do it for the particular case in which $I = 3$, $v(1) = v(2) = v(3) = 0$, and $0 \leq v(S) \leq v(I)$, for any group S of two agents.

Given a utility vector $u = (u_1, u_2, u_3) \geq 0$ and an $S \subset I$ the *excess* of S at u is $e(u, S) = v(S) - \sum_{i \in S} u_i$. We define the *first maximum excess* as $m_1(u) = \max \{e(u, S): 1 \leq \#S \leq 2\}$. Choose a two-agent coalition S such that $m_1(u) = e(u, S)$. Then we define the *second maximum excess* as $m_2(u) = \max \{e(u, S'): 1 \leq \#S' \leq 2 \text{ and } S' \neq S\}$.

We say that an exactly feasible [i.e., $\sum_{i \in I} u_i = v(I)$] utility profile $u = (u_1, u_2, u_3) \geq 0$ is in the nucleolus if for any other such profile u' we have either $m_1(u) < m_1(u')$ or $m_1(u) = m_1(u')$ and $m_2(u) \leq m_2(u')$.

(a) Show that if $u = (u_1, u_2, u_3)$ is in the nucleolus then either the three excesses for two-agent coalitions are identical or two are identical and the third is larger.

(b) Show that there is one and only one utility profile in the nucleolus. [Hint: Argue first that there is a two-agent coalition S such that $e(u, S) = m_1(u)$ for every profile in the nucleolus.] From now on we refer to this profile as the *nucleolus solution*.

(c) Argue that the nucleolus solution is symmetric.

- (d) Suppose that agent 1 is a dummy. Then $u_1 = 0$ at the nucleolus solution.
- (e) Suppose that $\frac{1}{2}v(I) \leq v(S)$ for any coalition S of two agents. Show then that at the nucleolus profile the three excesses for two-agent coalitions are identical.
- (f) Compute and compare the Shapley value and the nucleolus for the characteristic form: $v(1) = v(2) = v(3) = 0$, $v(\{1, 2\}) = v(\{1, 3\}) = 4$, $v(\{2, 3\}) = 5$, $v(I) = 6$.
- (g) Show that if the core is nonempty (see Appendix A to Chapter 18 for the definition of the core in this context) then the nucleolus utility profile belongs to the core.

22.F.6^B Consider a regulated firm that produces an output by means of a cost function $c(q)$. Assuming a quasilinear economy, the consumer surplus generated by q is $S(q)$.

(a) Suppose that $c(q)$ is strictly concave (i.e., strictly increasing returns to scale). Show that at the first-best price the firm will not cover costs. Conversely, for any q suppose that the price $p(q)$ is determined so that the cost is covered; that is, $p(q) = c(q)/q$. Show that if q is then determined so as to have $p(q) = S'(q)$, we will not reach the first-best optimum. Illustrate graphically.

(b) Suppose that the quantity produced, q , has to be determined under the constraint that with $p = S'(q)$ we have $pq \geq c(q)$. Solve this second-best welfare problem. Illustrate graphically.

(c) Interpret the units of output as “projects.” For any production decision q , what is the cost allocation suggested by the Shapley value?

22.F.7^C This exercise is similar to Exercise 22.F.6, except that the firm now produces two outputs under the separable cost functions $c_1(q_1)$, $c_2(q_2)$. The surplus $S_1(q_1) + S_2(q_2)$ is also separable.

(a) The second-best problem [first studied by Boiteux (1956)] is now richer than in Exercise 22.F.6. Suppose that the quantities q_1 , q_2 have to be determined so that with $p_1 = S'_1(q_1)$ and $p_2 = S'_2(q_2)$ we have $p_1q_1 + p_2q_2 \geq c_1(q_1) + c_2(q_2)$ (equivalently, at the chosen prices demand must be served and cost covered). Derive first-order conditions for this problem. Make them as similar as possible to the Ramsey formula of Example 22.B.2.

(b) (Harder) Interpret the units of outputs as projects. Suppose that these units are very small, so that a given production decision $(q_1, q_2) \gg 0$ represents the implementation of many projects of each of the two types. Can you guess, given (q_1, q_2) , what is an approximate value for the cost allocation suggested by the Shapley value? [Hint: For most orderings of projects, any particular project will have preceding it an almost perfect sample of all the projects.]

(c) Suppose that for the productions $(\bar{q}_1, \bar{q}_2)$, the Shapley value cost allocation assigns cost per unit of c_1 and c_2 (note that “projects” of the same type receive the same cost imputation). Suppose also that $c_1 = \partial S_1(\bar{q}_1)/\partial q_1$ and $c_2 = \partial S_2(\bar{q}_2)/\partial q_2$. Interpret. Argue that, in general, these productions will not correspond to either the first-best or the second-best optima of the problem.

Incentives and Mechanism Design

23.A Introduction

In Chapter 21, we studied how individual preferences might be aggregated into social preferences and ultimately into a collective decision. However, an important feature of many settings in which collective decisions must be made is that individuals' actual preferences are not publicly observable. As a result, in one way or another, individuals must be relied upon to reveal this information.

In this chapter, we study how this information can be elicited, and the extent to which the information revelation problem constrains the ways in which social decisions can respond to individual preferences. This topic is known as the *mechanism design problem*.

Mechanism design has many important applications throughout economics. The design of voting procedures, the writing of contracts among parties who will come to have private information, and the construction of procedures for deciding upon public projects or environmental standards are all examples.¹

The chapter is organized as follows. In Section 23.B, we introduce the mechanism design problem. We begin by illustrating the difficulties introduced by the need to elicit agents' preferences. We also define and discuss the concepts of *social choice functions* (already introduced in Section 21.E), *ex post efficiency*, *mechanisms*, *implementation*, *direct revelation mechanisms*, and *truthful implementation*.

In Section 23.C, we identify the circumstances under which a social choice function can be implemented in *dominant strategy equilibria* when agents' preferences are private information. Our analysis begins with a formal statement and proof of the *revelation principle*, a result that tells us that we can restrict attention to direct revelation mechanisms that induce agents to truthfully reveal their preferences. Using this fact, we then study the constraints that the information revelation

1. Simple examples of the last two applications were encountered in Sections 14.C and 11.E, respectively.

problem puts on the set of implementable social choice functions. We first present the important *Gibbard–Satterthwaite theorem*, which provides a very negative conclusion for cases in which individual preferences can take unrestricted forms. In the rest of the section, we go on to study the special case of *quasilinear environments*, discussing in detail *Groves–Clarke mechanisms*.

In Section 23.D, we study implementation in *Bayesian Nash equilibria*. We begin by discussing the *expected externality mechanism* as an example of how the weaker Bayesian implementation concept can allow us to implement a wider range of social choice functions than is possible with dominant strategy implementation. We go on to provide a characterization of Bayesian implementable social choice functions for the case in which agents have quasilinear preferences that are linear in their type. As an application of this result, we prove the remarkable *revenue equivalence theorem* for auctions.

In Section 23.E, we consider the possibility that participation in a mechanism may be voluntary and study how the need to satisfy the resulting *participation constraints* limits the set of implementable social choice functions. Here we prove the important *Myerson–Satterthwaite theorem*, which shows that, under very general conditions, it is impossible to achieve ex post efficiency in bilateral trade settings when agents have private information and trade is voluntary.

In Section 23.F, we discuss the welfare comparison of mechanisms, defining the notions of *ex ante* and *interim incentive efficiency*, and providing several illustrations of the computation of welfare optimal Bayesian mechanisms.

Appendices A and B are devoted to, first, a discussion of the issue of multiple equilibria in mechanism design and, second, the issue of mechanism design when agents know each others' types but the mechanism designer does not (so-called *complete information environments*).

References for further reading are provided at the start of the various sections. We would be remiss, however, not to mention here two early seminal articles: Mirrlees (1971) and Hurwicz (1972).

23.B The Mechanism Design Problem

In this section, we provide an introduction to the *mechanism design problem* that we study in detail in the rest of the chapter.

To begin, consider a setting with I agents, indexed by $i = 1, \dots, I$. These agents must make a collective choice from some set X of possible alternatives. Prior to the choice, however, each agent i privately observes his preferences over the alternatives in X . Formally, we model this by supposing that agent i privately observes a parameter, or signal, θ_i that determines his preferences. We will often refer to θ_i as agent i 's *type*. The set of possible types for agent i is denoted Θ_i . Each agent i is assumed to be an expected utility maximizer, whose Bernoulli utility function when he is of type θ_i is $u_i(x, \theta_i)$. The ordinal preference relation over pairs of alternatives in X that is associated with utility function $u_i(x, \theta_i)$ is denoted $\succsim_i(\theta_i)$. Agent i 's set of possible preference relations over X is therefore given by

$$\mathcal{R}_i = \{\succsim_i: \succsim_i = \succsim_i(\theta_i) \text{ for some } \theta_i \in \Theta_i\}.$$

Note that because θ_i is observed only by agent i , in the language of Section 8.E

we are in a setting characterized by *incomplete information*. As in Section 8.E, we suppose that agents' types are drawn from a commonly known prior distribution. In particular, denoting a profile of the agents' types by $\theta = (\theta_1, \dots, \theta_I)$, the probability density over the possible realizations of $\theta \in \Theta_1 \times \dots \times \Theta_I$ is $\phi(\cdot)$. The probability density $\phi(\cdot)$ as well as the sets $\Theta_1, \dots, \Theta_I$ and the utility functions $u_i(\cdot, \theta_i)$ are assumed to be common knowledge among the agents, but the specific value of each agent i 's type is observed only by i .²

Because the agents' preferences depend on the realizations of $\theta = (\theta_1, \dots, \theta_I)$, the agents may want the collective decision to depend on θ . To capture this dependence formally, we introduce in Definition 23.B.1 the notion of a *social choice function*, a concept already discussed in Section 21.E.³

Definition 23.B.1: A *social choice function* is a function $f: \Theta_1 \times \dots \times \Theta_I \rightarrow X$ that, for each possible profile of the agents' types $(\theta_1, \dots, \theta_I)$, assigns a collective choice $f(\theta_1, \dots, \theta_I) \in X$.⁴

One desirable feature for a social choice function to satisfy is the property of *ex post efficiency* described in Definition 23.B.2.

Definition 23.B.2: The social choice function $f: \Theta_1 \times \dots \times \Theta_I \rightarrow X$ is *ex post efficient* (or *Paretian*) if for no profile $\theta = (\theta_1, \dots, \theta_I)$ is there an $x \in X$ such that $u_i(x, \theta_i) \geq u_i(f(\theta), \theta_i)$ for every i , and $u_i(x, \theta_i) > u_i(f(\theta), \theta_i)$ for some i .

Definition 23.B.2 says that a social welfare function is *ex post efficient* if it selects, for every profile $\theta = (\theta_1, \dots, \theta_I)$, an alternative $f(\theta) \in X$ that is Pareto optimal given the agents' utility functions $u_1(\cdot, \theta_1), \dots, u_I(\cdot, \theta_I)$.

The problem faced by the agents is that the θ_i 's are not publicly observable, and so for the social choice $f(\theta_1, \dots, \theta_I)$ to be chosen when the agents' types are $(\theta_1, \dots, \theta_I)$, each agent i must be relied upon to disclose his type θ_i . However, for a given social choice function $f(\cdot)$, an agent may not find it to be in his best interest to reveal this information truthfully. We illustrate this information revelation problem in Examples 23.B.1 through 23.B.4, which range from very abstract to more applied settings.

2. The formulation here is restrictive in one sense: in some settings of interest, agents' preferences over outcomes depend not only on their own observed signals but also on signals observed by others (e.g., agent i 's preferences over whether to hold a picnic indoors may depend on agent j 's knowledge of likely weather conditions). Through most of this chapter, we focus on the case in which an agent's preferences depend only on his own signal, known as the *private values* case. We generalize our analysis in Section 23.F.

3. In Section 21.E an agent's type was equivalent to his ordinal preferences over X , and so a social choice function was defined there simply as a mapping from $\mathcal{R}_1 \times \dots \times \mathcal{R}_I$ to X . Moreover, it was assumed there that for all i we have $\mathcal{R}_i = \mathcal{R}$, the set of all possible ordinal preference orderings on X .

4. Two points should be noted about this definition. First, it restricts attention to *deterministic* social choice functions. This is largely for expositional purposes; although much of the chapter considers deterministic social choice functions, in Sections 23.D to 23.F we allow social choice functions that assign *lotteries* over X . Second, as in Section 21.E, we limit our attention to single-valued choice functions.

we are in a setting characterized by *incomplete information*. As in Section 8.E, we suppose that agents' types are drawn from a commonly known prior distribution. In particular, denoting a profile of the agents' types by $\theta = (\theta_1, \dots, \theta_I)$, the probability density over the possible realizations of $\theta \in \Theta_1 \times \dots \times \Theta_I$ is $\phi(\cdot)$. The probability density $\phi(\cdot)$ as well as the sets $\Theta_1, \dots, \Theta_I$ and the utility functions $u_i(\cdot, \theta_i)$ are assumed to be common knowledge among the agents, but the specific value of each agent i 's type is observed only by i .²

Because the agents' preferences depend on the realizations of $\theta = (\theta_1, \dots, \theta_I)$, the agents may want the collective decision to depend on θ . To capture this dependence formally, we introduce in Definition 23.B.1 the notion of a *social choice function*, a concept already discussed in Section 21.E.³

Definition 23.B.1: A *social choice function* is a function $f: \Theta_1 \times \dots \times \Theta_I \rightarrow X$ that, for each possible profile of the agents' types $(\theta_1, \dots, \theta_I)$, assigns a collective choice $f(\theta_1, \dots, \theta_I) \in X$.⁴

One desirable feature for a social choice function to satisfy is the property of *ex post efficiency* described in Definition 23.B.2.

Definition 23.B.2: The social choice function $f: \Theta_1 \times \dots \times \Theta_I \rightarrow X$ is *ex post efficient* (or *Paretian*) if for no profile $\theta = (\theta_1, \dots, \theta_I)$ is there an $x \in X$ such that $u_i(x, \theta_i) \geq u_i(f(\theta), \theta_i)$ for every i , and $u_i(x, \theta_i) > u_i(f(\theta), \theta_i)$ for some i .

Definition 23.B.2 says that a social welfare function is *ex post efficient* if it selects, for every profile $\theta = (\theta_1, \dots, \theta_I)$, an alternative $f(\theta) \in X$ that is Pareto optimal given the agents' utility functions $u_1(\cdot, \theta_1), \dots, u_I(\cdot, \theta_I)$.

The problem faced by the agents is that the θ_i 's are not publicly observable, and so for the social choice $f(\theta_1, \dots, \theta_I)$ to be chosen when the agents' types are $(\theta_1, \dots, \theta_I)$, each agent i must be relied upon to disclose his type θ_i . However, for a given social choice function $f(\cdot)$, an agent may not find it to be in his best interest to reveal this information truthfully. We illustrate this information revelation problem in Examples 23.B.1 through 23.B.4, which range from very abstract to more applied settings.

2. The formulation here is restrictive in one sense: in some settings of interest, agents' preferences over outcomes depend not only on their own observed signals but also on signals observed by others (e.g., agent i 's preferences over whether to hold a picnic indoors may depend on agent j 's knowledge of likely weather conditions). Through most of this chapter, we focus on the case in which an agent's preferences depend only on his own signal, known as the *private values* case. We generalize our analysis in Section 23.F.

3. In Section 21.E an agent's type was equivalent to his ordinal preferences over X , and so a social choice function was defined there simply as a mapping from $\mathcal{R}_1 \times \dots \times \mathcal{R}_I$ to X . Moreover, it was assumed there that for all i we have $\mathcal{R}_i = \mathcal{R}$, the set of all possible ordinal preference orderings on X .

4. Two points should be noted about this definition. First, it restricts attention to *deterministic* social choice functions. This is largely for expositional purposes; although much of the chapter considers deterministic social choice functions, in Sections 23.D to 23.F we allow social choice functions that assign *lotteries* over X . Second, as in Section 21.E, we limit our attention to single-valued choice functions.

Example 23.B.1: An Abstract Social Choice Setting. In the most abstract case, we are given a set X and, for each agent i , a set $\mathcal{R}_i$ of possible rational preference orderings on X . To consider a very simple example, suppose that $X = \{x, y, z\}$ and that $I = 2$. Suppose also that agent 1 has one possible type, so that $\Theta_1 = \{\bar{\theta}_1\}$, and that agent 2 has two possible types, so that $\Theta_2 = \{\theta'_2, \theta''_2\}$. The agents' possible preference orderings $\mathcal{R}_1 = \{\succsim_1(\bar{\theta}_1)\}$ and $\mathcal{R}_2 = \{\succsim_2(\theta'_2), \succsim_2(\theta''_2)\}$ are given by

$\succsim_1(\bar{\theta}_1)$	$\succsim_2(\theta'_2)$	$\succsim_2(\theta''_2)$
x	z	y
y	y	x
z	x	z

[A higher positioned alternative is strictly preferred to a lower positioned one; so, for example, $x \succ_1(\bar{\theta}_1) y \succ_1(\bar{\theta}_1) z$.]

Now suppose that the agents wish to implement the ex post efficient social choice function $f(\cdot)$ with

$$f(\bar{\theta}_1, \theta'_2) = y \quad \text{and} \quad f(\bar{\theta}_1, \theta''_2) = x.$$

If so, then agent 2 must be relied upon to truthfully reveal his preferences. But it is apparent that he will not find it in his interest to do so: When $\theta_2 = \theta''_2$, agent 2 will wish to lie and claim that his type is θ'_2 .

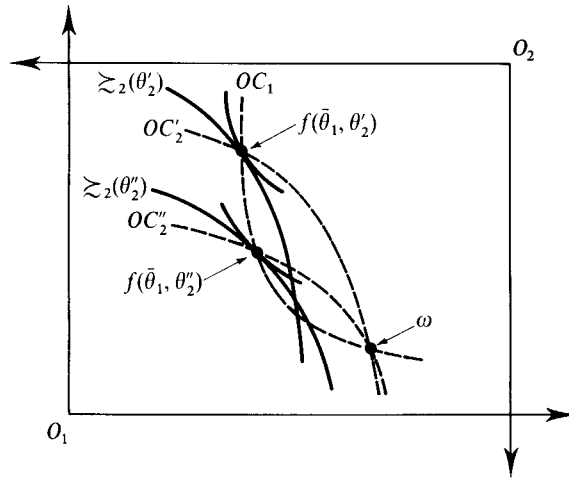
In abstract social choice settings, a case of central interest arises when $\mathcal{R}_i$ is, for each agent i , equal to $\mathcal{R}$, the set of all possible rational preference relations on X . In this case, an agent has many possible false claims that he can make and, intuitively, it may be very difficult for a social choice function always to induce the agents to reveal their preferences truthfully. We will see a formal illustration of this point in Section 23.C when we present the Gibbard–Satterthwaite theorem. ■

Example 23.B.2: A Pure Exchange Economy. Consider a pure exchange economy with L goods and I consumers in which agent i has consumption set $\mathbb{R}_+^L$ and endowment vector $\omega_i = (\omega_{i1}, \dots, \omega_{iL}) \gg 0$ (see Chapter 15). The set of alternatives is

$$X = \{(x_1, \dots, x_I) : x_i \in \mathbb{R}_+^L \text{ and } \sum_i x_{\ell i} \leq \sum_i \omega_{\ell i} \text{ for } \ell = 1, \dots, L\}.$$

In this setting it may be natural to suppose that $\mathcal{R}_i$, each consumer i 's set of possible preference relations over alternatives in X , is a subset of $\mathcal{R}_E$, the set of individualistic (i.e., depending on x_i only), monotone, and convex preference relations on X .

To consider a simple example, suppose that $I = 2$, that consumer 1 has only one possible type, so that $\Theta_1 = \{\bar{\theta}_1\}$ and $\mathcal{R}_1 = \{\succsim_1(\bar{\theta}_1)\}$, and that for consumer 2 we have $\mathcal{R}_2 = \mathcal{R}_E$. Imagine then that we try to implement a social choice function that, for each pair $(\succsim_1(\bar{\theta}_1), \succsim_2(\theta_2))$, chooses a Walrasian equilibrium allocation (note that this social choice function is ex post efficient). As Figure 23.B.1 illustrates, consumer 2 will not generally find it optimal to reveal his preferences truthfully. In the figure, $f(\bar{\theta}_1, \theta'_2)$ is the unique Walrasian equilibrium when preferences are $(\succsim_1(\bar{\theta}_1), \succsim_2(\theta'_2))$ [it is the unique intersection of the consumers' offer curves OC_1 and OC'_2 occurring at a point other than the endowment point]. However, by claiming that he has type θ''_2 , which has as its offer curve OC''_2 , consumer 2 can obtain the allocation $f(\bar{\theta}_1, \theta''_2)$ [the unique Walrasian equilibrium allocation when preferences

**Figure 23.B.1**

In the social choice function that selects a Walrasian equilibrium for each preference profile, agent 2 has an incentive to claim to be type θ_2'' when he is really type θ_2' .

are $(\succsim_1(\bar{\theta}_1), \succsim_2(\theta_2''))]$, an allocation that he strictly prefers to $f(\bar{\theta}_1, \theta_2')$ when his preferences are $\succsim_2(\theta_2')$. ■

Example 23.B.3: A Public Project. Consider a situation in which I agents must decide whether to undertake a public project, such as building a bridge, whose cost must be funded by the agents themselves. An outcome is a vector $x = (k, t_1, \dots, t_I)$, where $k \in \{0, 1\}$ is the decision whether to build the bridge ($k = 1$ if the bridge is built, and $k = 0$ if not), and $t_i \in \mathbb{R}$ is a monetary transfer to (or from, if $t_i < 0$) agent i . The cost of the project is $c \geq 0$ and so the set of feasible alternatives for the I agents is

$$X = \{(k, t_1, \dots, t_I) : k \in \{0, 1\}, t_i \in \mathbb{R} \text{ for all } i, \text{ and } \sum_i t_i \leq -ck\}.$$

The constraint $\sum_i t_i \leq -ck$ reflects the fact that there is no source of outside funding for the agents (so that we must have $c + \sum_i t_i \leq 0$ if $k = 1$, and $\sum_i t_i \leq 0$ if $k = 0$). We assume that type θ_i 's Bernoulli utility function has the quasilinear form

$$u_i(x, \theta_i) = \theta_i k + (\bar{m}_i + t_i),$$

where $\bar{m}_i$ is agent i 's initial endowment of the numeraire ("money") and $\theta_i \in \mathbb{R}$. We can then interpret θ_i as agent i 's willingness to pay for the bridge.

In this context, the social choice function $f(\theta) = (k(\theta), t_1(\theta), \dots, t_I(\theta))$ is ex post efficient if, for all θ ,

$$k(\theta) = \begin{cases} 1 & \text{if } \sum_i \theta_i \geq c, \\ 0 & \text{otherwise,} \end{cases} \quad (23.B.1)$$

and

$$\sum_i t_i(\theta) = -ck(\theta). \quad (23.B.2)$$

Suppose that the agents wish to implement a social choice function that satisfies (23.B.1) and (23.B.2) and in which an egalitarian contribution rule is followed, that is, in which $t_i(\theta) = -(c/I)k(\theta)$. To consider a simple example, suppose that $\Theta_i = \{\bar{\theta}_i\}$ for $i \neq 1$ (so that all agents other than agent 1 have preferences that are known) and

$\Theta_1 = [0, \infty)$. Suppose also that $c > \sum_{i \neq 1} \bar{\theta}_i > c(I-1)/I$. These inequalities imply, first, that with this social choice function agent 1's type is critical for whether the bridge is built (if $\theta_1 \geq c - \sum_{i \neq 1} \bar{\theta}_i$ it is; if $\theta_1 < c - \sum_{i \neq 1} \bar{\theta}_i$ it is not), and that the sum of the utilities of agents 2, ..., I is strictly greater if the bridge is built under this egalitarian contribution rule than if it is not built [since $\sum_{i \neq 1} \bar{\theta}_i - c(I-1)/I > 0$].

Let us examine agent 1's incentives for truthfully revealing his type when $\theta_1 = c - \sum_{i \neq 1} \bar{\theta}_i + \varepsilon$ for $\varepsilon > 0$. If agent 1 reveals his true preferences, the bridge will be built because

$$\left(c - \sum_{i \neq 1} \bar{\theta}_i + \varepsilon\right) + \sum_{i \neq 1} \bar{\theta}_i > c.$$

Agent 1's utility in this case is

$$\begin{aligned} \theta_1 + \bar{m}_1 - \frac{c}{I} &= \left(c - \sum_{i \neq 1} \bar{\theta}_i + \varepsilon\right) + \bar{m}_1 - \frac{c}{I} \\ &= \left(\frac{c(I-1)}{I} - \sum_{i \neq 1} \bar{\theta}_i + \varepsilon\right) + \bar{m}_1. \end{aligned}$$

But, for $\varepsilon > 0$ small enough, this is less than $\bar{m}_1$, which is agent 1's utility if he instead claims that $\theta_1 = 0$, a claim that results in the bridge not being built. Thus, agent 1 will prefer not to tell the truth. Intuitively, under this allocation rule, when agent 1 causes the bridge to be built he has a positive externality on the other agents (in the aggregate). Because he fails to internalize this effect, he has an incentive to understate his benefit from the project. ■

Example 23.B.4: Allocation of a Single Unit of an Indivisible Private Good. Consider a setting in which there is a single unit of an indivisible private good to be allocated to one of I agents. Monetary transfers can also be made. An outcome here may be represented by a vector $x = (y_1, \dots, y_I, t_1, \dots, t_I)$, where $y_i = 1$ if agent i gets the good, $y_i = 0$ if agent i does not get the good, and t_i is the monetary transfer received by agent i . The set of feasible alternatives is then

$$X = \{(y_1, \dots, y_I, t_1, \dots, t_I) : y_i \in \{0, 1\} \text{ and } t_i \in \mathbb{R} \text{ for all } i, \sum_i y_i = 1, \text{ and } \sum_i t_i \leq 0\}.$$

We suppose that type θ_i 's Bernoulli utility function takes the quasilinear form

$$u_i(x, \theta_i) = \theta_i y_i + (\bar{m}_i + t_i),$$

where $\bar{m}_i$ is once again agent i 's initial endowment of the numeraire ("money"). Here $\theta_i \in \mathbb{R}$ can be viewed as agent i 's valuation of the good, and we take the set of possible valuations for agent i to be $\Theta_i = [\underline{\theta}_i, \bar{\theta}_i] \subset \mathbb{R}$.

In this situation, a social choice function $f(\theta) = (y_1(\theta), \dots, y_I(\theta), t_1(\theta), \dots, t_I(\theta))$ is ex post efficient if it always allocates the good to the agent who has the highest valuation (or to one of them if there are several) and if it involves no waste of the numeraire; that is, if for all $\theta = (\theta_1, \dots, \theta_I) \in \Theta_1 \times \dots \times \Theta_I$,

$$y_i(\theta)(\theta_i - \text{Max}\{\theta_1, \dots, \theta_I\}) = 0 \quad \text{for all } i$$

and

$$\sum_i t_i(\theta) = 0.$$

Two special cases that have received a great deal of attention in the literature deserve mention. The first is the case of *bilateral trade*. In this case we have $I = 2$; agent 1 is interpreted as the initial owner of the good (the “seller”), and agent 2 is the potential purchaser of the good (the “buyer”). When $\underline{\theta}_2 > \bar{\theta}_1$ there are certain to be gains from trade regardless of the realizations of θ_1 and θ_2 ; when $\underline{\theta}_1 > \bar{\theta}_2$ there are certain to be no gains from trade; finally, if $\underline{\theta}_2 < \bar{\theta}_1$ and $\underline{\theta}_1 < \bar{\theta}_2$ then there may or may not be gains from trade, depending on the realization of θ .

The second special case is the *auction* setting. Here, one agent, whom we shall designate as agent 0, is interpreted as the seller of the good (the “auctioneer”) and is assumed to derive no value from it (more generally, the seller might have a known value $\theta_0 = \bar{\theta}_0$ different from zero). The other agents, $1, \dots, I$, are potential buyers (the “bidders”).⁵

To illustrate the problem with information revelation in this example, consider an auction setting with two buyers ($I = 2$). In the previous examples, we simplified the discussion of information revelation by assuming that only one agent has more than one possible type. We now suppose instead that both buyers’ (privately observed) valuations θ_i are drawn independently from the uniform distribution on $[0, 1]$ and that this fact is common knowledge among the agents. Consider the social choice function $f(\theta) = (y_0(\theta), y_1(\theta), y_2(\theta), t_0(\theta), t_1(\theta), t_2(\theta))$ in which

$$y_1(\theta) = 1 \quad \text{if } \theta_1 \geq \theta_2; \quad = 0 \text{ if } \theta_1 < \theta_2 \quad (23.B.3)$$

$$y_2(\theta) = 1 \quad \text{if } \theta_1 < \theta_2; \quad = 0 \text{ if } \theta_1 \geq \theta_2 \quad (23.B.4)$$

$$y_0(\theta) = 0 \quad \text{for all } \theta \quad (23.B.5)$$

$$t_1(\theta) = -\theta_1 y_1(\theta) \quad (23.B.6)$$

$$t_2(\theta) = -\theta_2 y_2(\theta) \quad (23.B.7)$$

$$t_0(\theta) = -(t_1(\theta) + t_2(\theta)). \quad (23.B.8)$$

In this social choice function, the seller gives the good to the buyer with the highest valuation (to buyer 1 if there is a tie) and this buyer gives the seller a payment equal to his valuation (the other, low-valuation buyer makes no transfer payment to the seller). Note that $f(\cdot)$ is not only ex post efficient but also is very attractive for the seller: if $f(\cdot)$ can be implemented, the seller will capture all of the consumption benefits that are generated by the good.

Suppose we try to implement this social choice function. Assume that the buyers are expected utility maximizers. We now ask: If buyer 2 always announces his true value, will buyer 1 find it optimal to do the same? For each value of θ_1 , buyer 1’s problem is to choose the valuation to announce, say $\hat{\theta}_1$, so as to solve

$$\text{Max}_{\hat{\theta}_1} (\theta_1 - \hat{\theta}_1) \text{Prob}(\theta_2 \leq \hat{\theta}_1)$$

or

$$\text{Max}_{\hat{\theta}_1} (\theta_1 - \hat{\theta}_1) \hat{\theta}_1.$$

5. Note that, for ease of notation, we take there to be $I + 1$ agents in the auction setting.

The solution to this problem has buyer 1 set $\hat{\theta}_1 = \theta_1/2$. We see then that if buyer 2 always tells the truth, truth telling is *not* optimal for buyer 1. A similar point applies to buyer 2. Intuitively, for this social choice function, a buyer has an incentive to understate his valuation so as to lower the transfer he must make in the event that he has the highest announced valuation and gets the good. The cost to him of doing this is that he gets the good less often, but this is a cost worth incurring to at least some degree.⁶ Thus, we again see that there may be a problem in implementing certain social choice functions in settings in which information is privately held. (For a similar point in the bilateral trade context, see Exercise 23.B.2.)

Although buyers have an incentive to lie given the social choice function described in (23.B.3) to (23.B.8), this is *not* true of *all* social choice functions in this auction setting. To see this point, suppose we try to implement the social choice function $\tilde{f}(\cdot)$ that has the same allocation rule as that above [i.e., in which the functions $y_i(\cdot)$ for $i = 0, 1, 2$ are the same as those described in (23.B.3) to (23.B.5)] but instead has transfer functions

$$t_1(\theta) = -\theta_2 y_1(\theta)$$

$$t_2(\theta) = -\theta_1 y_2(\theta)$$

$$t_0(\theta) = -(t_1(\theta) + t_2(\theta)).$$

In this social choice function, instead of buyer i paying the seller an amount equal to his own valuation θ_i if he wins the object, he now pays θ_j , where $j \neq i$; that is, he pays an amount equal to the *second-highest valuation*. Consider buyer 1's incentives for truth telling now. If buyer 2 announces his valuation to be $\hat{\theta}_2 \leq \theta_1$, buyer 1 can receive a utility of $(\theta_1 - \hat{\theta}_2) \geq 0$ by truthfully announcing that his valuation is θ_1 . For any other announcement, buyer 1's resulting utility is either the same (if he announces a valuation of at least $\hat{\theta}_2$) or zero (if he announces a valuation below $\hat{\theta}_2$). So if $\hat{\theta}_2 \leq \theta_1$, announcing the truth is weakly best for buyer 1. On the other hand, if buyer 2's announced valuation is $\hat{\theta}_2 > \theta_1$, then buyer 1's utility is 0 if he reveals his true valuation. However, buyer 1 can receive only a negative utility by making a false claim that gets him the good (a claim that his valuation is at least $\hat{\theta}_2$). We conclude that truth telling is optimal for buyer 1 regardless of what buyer 2 announces. Formally, in the language of the theory of games, truth telling is a weakly dominant strategy for buyer 1 (see Section 8.B). A similar conclusion follows for buyer 2. Thus, this social choice function *is* implementable even though the buyers' valuations are private information: it suffices to simply ask each buyer to report his type, and then to choose $\tilde{f}(\theta)$.⁷ ■

Examples 23.B.1 to 23.B.4 suggest that when agents' types are privately observed the information revelation problem may constrain the set of social choice functions that can be successfully implemented. With these examples as motivation, we can now pose the central question that is our focus in this chapter: *What social choice functions can be implemented when agents' types are private information?*

6. This trade-off is similar to that faced by a monopolist (see Section 12.B): when the monopolist raises his price, he lowers his sales but makes more on his remaining sales.

7. For other examples of implementable social choice functions, see Exercise 23.B.1.

To answer this question, we need in principle to begin by thinking of all the possible ways in which a social choice function might be implemented. In the above examples we have implicitly imagined a very simple scenario in which each agent i is asked to directly reveal θ_i and then, given the announcements $(\hat{\theta}_1, \dots, \hat{\theta}_I)$, the alternative $f(\hat{\theta}_1, \dots, \hat{\theta}_I) \in X$ is chosen. But this is not the only way a social choice function might be implemented. In particular, a given social choice function might be *indirectly* implemented by having the agents interact through some type of institution in which there are rules governing the actions the agents may take and how these actions translate into a social outcome. To illustrate this point, Examples 23.B.5 and 23.B.6 study two commonly used auction institutions.

Example 23.B.5: First-Price Sealed-Bid Auction. Consider again the auction setting introduced in Example 23.B.4. In a *first-price sealed-bid auction* each potential buyer i is allowed to submit a sealed bid, $b_i \geq 0$. The bids are then opened and the buyer with the highest bid gets the good and pays an amount equal to his bid to the seller.⁸

To be specific, consider again the case where there are two potential buyers ($I = 2$) and each θ_i is independently drawn from the uniform distribution on $[0, 1]$. We will look for an equilibrium in which each buyer's strategy $b_i(\cdot)$ takes the form $b_i(\theta_i) = \alpha_i \theta_i$ for $\alpha_i \in [0, 1]$. Suppose that buyer 2's strategy has this form, and consider buyer 1's problem. For each θ_1 he wants to solve

$$\text{Max}_{b_1 \geq 0} (\theta_1 - b_1) \text{Prob}(b_2(\theta_2) \leq b_1).$$

Since buyer 2's highest possible bid is α_2 (he submits a bid of α_2 when $\theta_2 = 1$), it is evident that buyer 1 should never bid more than α_2 . Moreover, since θ_2 is uniformly distributed on $[0, 1]$ and $b_2(\theta_2) \leq b_1$ if and only if $\theta_2 \leq (b_1/\alpha_2)$, we can write buyer 1's problem as

$$\text{Max}_{b_1 \in [0, \alpha_2]} (\theta_1 - b_1)(b_1/\alpha_2).$$

The solution to this problem is

$$b_1(\theta_1) = \begin{cases} \frac{1}{2}\theta_1 & \text{if } \frac{1}{2}\theta_1 \leq \alpha_2, \\ \alpha_2 & \text{if } \frac{1}{2}\theta_1 > \alpha_2. \end{cases}$$

By similar reasoning,

$$b_2(\theta_2) = \begin{cases} \frac{1}{2}\theta_2 & \text{if } \frac{1}{2}\theta_2 \leq \alpha_1, \\ \alpha_1 & \text{if } \frac{1}{2}\theta_2 > \alpha_1. \end{cases}$$

Letting $\alpha_1 = \alpha_2 = \frac{1}{2}$, we see that the strategies $b_i(\theta_i) = \frac{1}{2}\theta_i$ for $i = 1, 2$ constitute a Bayesian Nash equilibrium for this auction. Thus, there is a Bayesian Nash equilibrium of this first-price sealed-bid auction that indirectly yields the outcomes specified by the social choice function $f(\theta) = (y_0(\theta), y_1(\theta), y_2(\theta), t_0(\theta), t_1(\theta), t_2(\theta))$

8. If there are several highest bids, we suppose that the lowest numbered of these bidders gets the good. We could equally well randomize among the highest bidders if there are more than one, but this would require that we expand the set of alternatives to $\Delta(X)$, the set of all lotteries over X . In fact, we do precisely this when we study auctions in Sections 23.D and 23.F.

in which

$$y_1(\theta) = 1 \quad \text{if } \theta_1 \geq \theta_2; \quad = 0 \text{ if } \theta_1 < \theta_2 \quad (23.B.9)$$

$$y_2(\theta) = 1 \quad \text{if } \theta_1 < \theta_2; \quad = 0 \text{ if } \theta_1 \geq \theta_2 \quad (23.B.10)$$

$$y_0(\theta) = 0 \quad \text{for all } \theta \quad (23.B.11)$$

$$t_1(\theta) = -\frac{1}{2}\theta_1 y_1(\theta) \quad (23.B.12)$$

$$t_2(\theta) = -\frac{1}{2}\theta_2 y_2(\theta) \quad (23.B.13)$$

$$t_0(\theta) = -(t_1(\theta) + t_2(\theta)). \quad (23.B.14)$$

■

Example 23.B.6: Second-Price Sealed-Bid Auction.⁹ Once again, consider the auction setting described in Example 23.B.4. In a *second-price sealed-bid auction*, each potential buyer i is allowed to submit a sealed bid, $b_i \geq 0$. The bids are then opened and the buyer with the highest bid gets the good, but now he pays the seller an amount equal to the *second-highest* bid.¹⁰

By reasoning that parallels that at the end of Example 23.B.4, the strategy $b_i(\theta_i) = \theta_i$ for all $\theta_i \in [0, 1]$ is a weakly dominant strategy for each buyer i (see Exercise 23.B.3). Thus, when $I = 2$ the second-price sealed-bid auction implements the social choice function $f(\theta) = (y_0(\theta), y_1(\theta), y_2(\theta), t_0(\theta), t_1(\theta), t_2(\theta))$ in which

$$y_1(\theta) = 1 \quad \text{if } \theta_1 \geq \theta_2; \quad = 0 \text{ if } \theta_1 < \theta_2$$

$$y_2(\theta) = 1 \quad \text{if } \theta_1 < \theta_2; \quad = 0 \text{ if } \theta_1 \geq \theta_2$$

$$y_0(\theta) = 0 \quad \text{for all } \theta$$

$$t_1(\theta) = -\theta_2 y_1(\theta)$$

$$t_2(\theta) = -\theta_1 y_2(\theta)$$

$$t_0(\theta) = -(t_1(\theta) + t_2(\theta)). \quad \blacksquare$$

Examples 23.B.5 and 23.B.6 illustrate that, as a general matter, we need to consider not only the possibility of directly implementing social choice functions by asking agents to reveal their types but also their indirect implementation through the design of institutions in which the agents interact. The formal representation of such an institution is known as a *mechanism*.

Definition 23.B.3: A *mechanism* $\Gamma = (S_1, \dots, S_I, g(\cdot))$ is a collection of I strategy sets $(S_1, \dots, S_I)$ and an outcome function $g: S_1 \times \dots \times S_I \rightarrow X$.

A mechanism can be viewed as an institution with rules governing the procedure for making the collective choice. The allowed actions of each agent i are summarized by the strategy set S_i , and the rule for how agents' actions get turned into a social choice is given by the outcome function $g(\cdot)$.

Formally, the mechanism Γ combined with possible types $(\Theta_1, \dots, \Theta_I)$, probability density $\phi(\cdot)$, and Bernoulli utility functions $(u_1(\cdot), \dots, u_I(\cdot))$ defines

9. This auction is also called a *Vickrey auction*, after Vickrey (1961).

10. If there is more than one highest bid, we again select the lowest-numbered of these bidders.

a Bayesian game of incomplete information. That is, letting $\tilde{u}_i(s_1, \dots, s_I, \theta_i) = u_i(g(s_1, \dots, s_I), \theta_i)$, the game

$$[I, \{S_i\}, \{\tilde{u}_i(\cdot)\}, \Theta_1 \times \dots \times \Theta_I, \phi(\cdot)]$$

is exactly the type of Bayesian game studied in Section 8.E. Note that a mechanism could in principle be a complex dynamic procedure, in which case the elements of the strategy sets S_i would consist of contingent plans of action (see Chapter 7).¹¹

For the auction setting, the first-price sealed-bid auction is the mechanism in which $S_i = \mathbb{R}_+$ for all i and, given the bids $(b_1, \dots, b_I) \in \mathbb{R}_+^I$, the outcome function $g(b_1, \dots, b_I) = (\{y_i(b_1, \dots, b_I)\}_{i=1}^I, \{t_i(b_1, \dots, b_I)\}_{i=1}^I)$ is such that

$$y_i(b_1, \dots, b_I) = 1 \quad \text{if and only if} \quad i = \text{Min}\{j: b_j = \text{Max}\{b_1, \dots, b_I\}\},$$

$$t_i(b_1, \dots, b_I) = -b_i y_i(b_1, \dots, b_I).$$

In the second-price sealed-bid auction, on the other hand, we have the same strategy sets and functions $y_i(\cdot)$, but instead $t_i(b_1, \dots, b_I) = -\text{Max}\{b_j: j \neq i\} y_i(b_1, \dots, b_I)$.

A strategy for agent i in the game of incomplete information created by a mechanism Γ is a function $s_i: \Theta_i \rightarrow S_i$ giving agent i 's choice from S_i for each possible type in Θ_i that he might have. Loosely put, we say that a mechanism *implements* social choice function $f(\cdot)$ if there is an equilibrium of the game induced by the mechanism that yields the same outcomes as $f(\cdot)$ for each possible profile of types $\theta = (\theta_1, \dots, \theta_I)$. This is stated formally in Definition 23.B.4.

Definition 23.B.4: The mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ *implements* social choice function $f(\cdot)$ if there is an equilibrium strategy profile $(s_1^*(\cdot), \dots, s_I^*(\cdot))$ of the game induced by Γ such that $g(s_1^*(\theta_1), \dots, s_I^*(\theta_I)) = f(\theta_1, \dots, \theta_I)$ for all $(\theta_1, \dots, \theta_I) \in \Theta_1 \times \dots \times \Theta_I$.

Note, however, that we have not specified in Definition 23.B.4 exactly what we mean by an “equilibrium”. This is because, as we have seen in Part II, there is no single equilibrium concept that is universally agreed upon as *the* appropriate solution concept for games. As a result, the mechanism design literature has investigated the implementation question for a variety of solution concepts. In Sections 23.C and 23.D we focus on two central solution concepts: dominant strategy equilibrium and Bayesian Nash equilibrium.¹²

Note also that the notion of implementation that we have adopted in Definition 23.B.4 is in one sense a weak one: in particular, the mechanism Γ may have *more than one equilibrium*, but Definition 23.B.4 requires only that *one of them* induce outcomes in accord with $f(\cdot)$. Implicitly, then, Definition 23.B.4 assumes that, if multiple equilibria exist, the agents will play the equilibrium that the mechanism designer wants. Throughout the chapter we shall stick to this notion of implementation. Appendix A is devoted to a further discussion of this issue.

11. Note also that we are representing the game created by a mechanism using its normal form. For all the analysis that follows in the text this will be sufficient. In Appendix B, however, we consider a case where the extensive form representation is used.

12. Appendix B considers several other equilibrium concepts in the special context of *complete information* settings in which the players observe each others' types.

The identification of all social choice functions that are implementable may seem like a daunting task because, in principle, it appears that we need to consider all possible mechanisms—a very large set. Fortunately, an important result known as the *revelation principle* (to be formally stated and proven in Sections 23.C and 23.D) tells us that we can often restrict attention to the very simple type of mechanisms that we were implicitly considering at the outset, that is, mechanisms in which each agent is asked to reveal his type, and given the announcements $(\hat{\theta}_1, \dots, \hat{\theta}_I)$, the alternative chosen is $f(\hat{\theta}_1, \dots, \hat{\theta}_I) \in X$.¹³ These are known as *direct revelation mechanisms*, and formally constitute a special case of the mechanisms of Definition 23.B.3.

Definition 23.B.5: A *direct revelation mechanism* is a mechanism in which $S_i = \Theta_i$ for all i and $g(\theta) = f(\theta)$ for all $\theta \in \Theta_1 \times \dots \times \Theta_I$.

Moreover, as we shall see, the revelation principle also tells us that we can further restrict our attention to direct revelation mechanisms in which *truth telling is an optimal strategy for each agent*. This fact motivates the notion of *truthful implementation* that we introduce in Definition 23.B.6 (we are again purposely vague in the definition about the equilibrium concept we wish to employ).

Definition 23.B.6: The social choice function $f(\cdot)$ is *truthfully implementable* (or *incentive compatible*) if the direct revelation mechanism $\Gamma = (\Theta_1, \dots, \Theta_I, f(\cdot))$ has an equilibrium $(s_1^*(\cdot), \dots, s_I^*(\cdot))$ in which $s_i^*(\theta_i) = \theta_i$ for all $\theta_i \in \Theta_i$ and all $i = 1, \dots, I$; that is, if truth telling by each agent i constitutes an equilibrium of $\Gamma = (\Theta_1, \dots, \Theta_I, f(\cdot))$.

To offer a hint as to why we may be able to restrict attention to direct revelation mechanisms that induce truth telling, we briefly verify that the social choice functions that are implemented indirectly through the first-price and second-price sealed-bid auctions of Examples 23.B.5 and 23.B.6 can also be truthfully implemented using a direct revelation mechanism. In fact, for the second-price sealed-bid auction of Example 23.B.6 we have already seen this fact, because the social choice function implemented by the second-price auction is exactly the social choice function that we studied at the end of Example 23.B.4 in which truth telling is a weakly dominant strategy for both buyers. Example 23.B.7 considers the first-price sealed-bid auction.

Example 23.B.7: Truthful Implementation of the Social Choice Function Implemented by the First-Price Sealed-Bid Auction. When facing the direct revelation mechanism $(\Theta_1, \dots, \Theta_I, f(\cdot))$ with $f(\theta) = (y_0(\theta), y_1(\theta), y_2(\theta), t_0(\theta), t_1(\theta), t_2(\theta))$ satisfying (23.B.9) to (23.B.14), buyer 1's optimal announcement $\hat{\theta}_1$ when he has type θ_1 solves

$$\text{Max}_{\hat{\theta}_1} (\theta_1 - \tfrac{1}{2}\hat{\theta}_1) \text{Prob}(\theta_2 \leq \hat{\theta}_1)$$

or

$$\text{Max}_{\hat{\theta}_1} (\theta_1 - \tfrac{1}{2}\hat{\theta}_1)\hat{\theta}_1.$$

13. Some early versions of the revelation principle were derived by Gibbard (1973), Green and Laffont (1977), Myerson (1979), and Dasgupta, Hammond and Maskin (1979).

The first-order condition for this problem gives $\hat{\theta}_1 = \theta_1$. So truth telling is buyer 1's optimal strategy given that buyer 2 always tells the truth. A similar conclusion follows for buyer 2. Thus, the social choice function implemented by the first-price sealed-bid auction (in a Bayesian Nash equilibrium) can also be truthfully implemented (in a Bayesian Nash equilibrium) through a direct revelation mechanism. That is, the social choice function (23.B.9) to (23.B.14) is incentive compatible. ■

Because of the revelation principle, when we explore in Sections 23.C and 23.D the constraints that incomplete information about types puts on the set of implementable social choice functions, we will be able to restrict our analysis to identifying those social choice functions that can be truthfully implemented.

Finally, we note that, in some applications, participation in the mechanism may be *voluntary*, and so a social choice function must not only induce truthful revelation of information but must also satisfy certain *participation* (or *individual rationality*) constraints if it is to be successfully implemented. In Sections 23.C and 23.D, however, we shall abstract from issues of participation to focus exclusively on the information revelation problem. We introduce participation constraints in Section 23.E.

23.C Dominant Strategy Implementation

In this section, we study implementation in *dominant strategies*.¹⁴ Throughout we follow the notation introduced in Section 23.B: The vector of agents' types $\theta = (\theta_1, \dots, \theta_I)$ is drawn from the set $\Theta = \Theta_1 \times \dots \times \Theta_I$ according to a probability density $\phi(\cdot)$, and agent i 's Bernoulli utility function over the alternatives in X given his type θ_i is $u_i(x, \theta_i)$. We also adopt the notational convention of writing $\theta_{-i} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_I)$, $\theta = (\theta_i, \theta_{-i})$, and $\Theta_{-i} = \Theta_1 \times \dots \times \Theta_{i-1} \times \Theta_{i+1} \times \dots \times \Theta_I$. A mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ is a collection of I sets $S_1, \dots, S_I$, each S_i containing agent i 's possible actions (or plans of action), and an outcome function $g: S \rightarrow X$, where $S = S_1 \times \dots \times S_I$. As discussed in Section 23.B, a mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ combined with possible types $(\Theta_1, \dots, \Theta_I)$, density $\phi(\cdot)$, and Bernoulli utility functions $(u_1(\cdot), \dots, u_I(\cdot))$ defines a Bayesian game of incomplete information (see Section 8.E). We will also often write $s_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_I)$, $s = (s_i, s_{-i})$, and $S_{-i} = S_1 \times \dots \times S_{i-1} \times S_{i+1} \times \dots \times S_I$.

Recall from Section 8.B that a strategy is a weakly dominant strategy for a player in a game if it gives him at least as large a payoff as any of his other possible strategies for every possible strategy that his rivals might play. In the present incomplete information environment, strategy $s_i: \Theta_i \rightarrow S_i$ is a weakly dominant strategy for agent i in mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ if, for all $\theta_i \in \Theta_i$ and all possible strategies for agents $j \neq i$, $s_{-i}(\cdot) = [s_1(\cdot), \dots, s_{i-1}(\cdot), s_{i+1}(\cdot), \dots, s_I(\cdot)]$,¹⁵

$$E_{\theta_{-i}}[u_i(g(s_i(\theta_i), s_{-i}(\theta_{-i})), \theta_i) | \theta_i] \geq E_{\theta_{-i}}[u_i(g(\hat{s}_i, s_{-i}(\theta_{-i})), \theta_i) | \theta_i] \quad \text{for all } \hat{s}_i \in S_i. \quad (23.C.1)$$

Condition (23.C.1) holding for all $s_{-i}(\cdot)$ and θ_i is equivalent to the condition that,

14. Good sources for further reading on the subject of this section are Dasgupta, Hammond and Maskin (1979) and Green and Laffont (1979).

15. The expectation in (23.C.1) is taken over realizations of $\theta_{-i} \in \Theta_{-i}$.

for all $\theta_i \in \Theta_i$,

$$u_i(g(s_i(\theta_i), s_{-i}), \theta_i) \geq u_i(g(\hat{s}_i, s_{-i}), \theta_i) \quad (23.C.2)$$

for all $\hat{s}_i \in S_i$ and all $s_{-i} \in S_{-i}$.¹⁶ This leads to Definition 23.C.1.

Definition 23.C.1: The strategy profile $s^*(\cdot) = (s_1^*(\cdot), \dots, s_I^*(\cdot))$ is a *dominant strategy equilibrium* of mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ if, for all i and all $\theta_i \in \Theta_i$,

$$u_i(g(s_i^*(\theta_i), s_{-i}), \theta_i) \geq u_i(g(s'_i, s_{-i}), \theta_i)$$

for all $s'_i \in S_i$ and all $s_{-i} \in S_{-i}$.

We now specialize Definition 23.B.4 to the notion of dominant strategy equilibrium.

Definition 23.C.2: The mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ implements the social choice function $f(\cdot)$ in dominant strategies if there exists a dominant strategy equilibrium of Γ , $s^*(\cdot) = (s_1^*(\cdot), \dots, s_I^*(\cdot))$, such that $g(s^*(\theta)) = f(\theta)$ for all $\theta \in \Theta$.

The concept of dominant strategy implementation is of special interest because if we can find a mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ that implements $f(\cdot)$ in dominant strategies, then this mechanism implements $f(\cdot)$ in a very strong and robust way. This is true in several senses. First, we can feel fairly confident that a rational agent who has a (weakly) dominant strategy will indeed play it.¹⁷ Unlike the equilibrium strategies in Nash-related equilibrium concepts, a player need not correctly forecast his opponents' play to justify his play of a dominant strategy. Second, although we have assumed that the agents know the probability density $\phi(\cdot)$ over realizations of the types $(\theta_1, \dots, \theta_I)$, and hence can deduce the correct conditional probability distribution over realizations of θ_{-i} , if Γ implements $f(\cdot)$ in dominant strategies this implementation will be robust even if agents have incorrect, and perhaps even contradictory, beliefs about this distribution. In particular, agent i 's beliefs regarding the distribution of θ_{-i} do not affect the dominance of his strategy $s_i^*(\cdot)$.¹⁸ Third, it follows that if Γ implements $f(\cdot)$ in dominant strategies then it does so regardless of the probability density $\phi(\cdot)$. Thus, the same mechanism can be used to implement $f(\cdot)$ for any $\phi(\cdot)$. One advantage of this is that if the mechanism designer is an outsider (say, the "government"), he need not know $\phi(\cdot)$ to successfully implement $f(\cdot)$.

As we noted in Section 23.B, to identify whether a particular social choice function $f(\cdot)$ is implementable, we need, in principle, to consider all possible mechanisms. Fortunately, it turns out that for dominant strategy implementation it suffices to ask

16. Condition (23.C.2) follows from (23.C.1) simply by setting $s_{-i}(\theta_{-i}) = s_{-i}$ for all $\theta_{-i} \in \Theta_{-i}$. To see that (23.C.2) implies (23.C.1), consider the case where S_{-i} is a finite set. Then, for any s_i ,

$$E_{\theta_{-i}}[u_i(g(s_i, s_{-i}(\theta_{-i})), \theta_i) | \theta_i] = \sum_{s_{-i} \in S_{-i}} \text{Prob}(s_{-i}(\theta_{-i}) = s_{-i}) u_i(g(s_i, s_{-i}), \theta_i).$$

Thus, (23.C.2) implies (23.C.1).

17. We leave aside the question of what might happen if an agent has several weakly dominant strategies. This is the issue of multiple equilibria that we discuss in Appendix A. Even so, we at least mention one conclusion from that discussion: The problem of multiple equilibria is relatively small when we are dealing with dominant strategy equilibrium.

18. In fact, the implementation of $f(\cdot)$ using Γ is also robust to substantial relaxations of the hypothesis that agents maximize expected utility.

whether a particular $f(\cdot)$ is truthfully implementable in the sense introduced in Definition 23.C.3.

Definition 23.C.3: The social choice function $f(\cdot)$ is *truthfully implementable in dominant strategies* (or *dominant strategy incentive compatible*, or *strategy-proof*, or *straightforward*) if $s_i^*(\theta_i) = \theta_i$ for all $\theta_i \in \Theta_i$ and $i = 1, \dots, I$ is a dominant strategy equilibrium of the direct revelation mechanism $\Gamma = (\Theta_1, \dots, \Theta_I, f(\cdot))$. That is, if for all i and all $\theta_i \in \Theta_i$,

$$u_i(f(\theta_i, \theta_{-i}), \theta_i) \geq u_i(f(\hat{\theta}_i, \theta_{-i}), \theta_i) \quad (23.C.3)$$

for all $\hat{\theta}_i \in \Theta_i$ and all $\theta_{-i} \in \Theta_{-i}$.

The ability to restrict our inquiry, without loss of generality, to the question of whether $f(\cdot)$ is truthfully implementable is a consequence of what is known as the *revelation principle for dominant strategies*.

Proposition 23.C.1: (The Revelation Principle for Dominant Strategies) Suppose that there exists a mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ that implements the social choice function $f(\cdot)$ in dominant strategies. Then $f(\cdot)$ is truthfully implementable in dominant strategies.

Proof: If $\Gamma = (S_1, \dots, S_I, g(\cdot))$ implements $f(\cdot)$ in dominant strategies, then there exists a profile of strategies $s^*(\cdot) = (s_1^*(\cdot), \dots, s_I^*(\cdot))$ such that $g(s^*(\theta)) = f(\theta)$ for all θ and, for all i and all $\theta_i \in \Theta_i$,

$$u_i(g(s_i^*(\theta_i), s_{-i}), \theta_i) \geq u_i(g(\hat{s}_i, s_{-i}), \theta_i) \quad (23.C.4)$$

for all $\hat{s}_i \in S_i$ and all $s_{-i} \in S_{-i}$. Condition (23.C.4) implies, in particular, that for all i and all $\theta_i \in \Theta_i$,

$$u_i(g(s_i^*(\theta_i), s_{-i}^*(\theta_{-i})), \theta_i) \geq u_i(g(s_i^*(\hat{\theta}_i), s_{-i}^*(\theta_{-i})), \theta_i) \quad (23.C.5)$$

for all $\hat{\theta}_i \in \Theta_i$ and all $\theta_{-i} \in \Theta_{-i}$. Since $g(s^*(\theta)) = f(\theta)$ for all θ , (23.C.5) means that, for all i and all $\theta_i \in \Theta_i$,

$$u_i(f(\theta_i, \theta_{-i}), \theta_i) \geq u_i(f(\hat{\theta}_i, \theta_{-i}), \theta_i)$$

for all $\hat{\theta}_i \in \Theta_i$ and all $\theta_{-i} \in \Theta_{-i}$. But, this is precisely condition (23.C.3), the condition for $f(\cdot)$ to be truthfully implementable in dominant strategies. ■

The intuitive idea behind the revelation principle for dominant strategies can be put as follows: Suppose that the indirect mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ implements $f(\cdot)$ in dominant strategies, and that in this indirect mechanism each agent i finds playing $s_i^*(\theta_i)$ when his type is θ_i better than playing any other $s_i \in S_i$ for any choices $s_{-i} \in S_{-i}$ by agents $j \neq i$. Now consider altering this mechanism simply by introducing a mediator who says to each agent i : “You tell me your type, and when you say your type is θ_i , I will play $s_i^*(\theta_i)$ for you.” Clearly, if $s_i^*(\theta_i)$ is agent i ’s optimal choice for each $\theta_i \in \Theta_i$ in the initial mechanism Γ for any strategies chosen by the other agents, then agent i will find telling the truth to be a dominant strategy in this new scheme. But this means that we have found a way to truthfully implement $f(\cdot)$.

The implication of the revelation principle is that to identify the set of social choice functions that are implementable in dominant strategies, we need only identify those that are truthfully implementable. In principle, for any $f(\cdot)$, this is just a matter of checking the inequalities (23.C.3).

The inequalities (23.C.3), which are necessary and sufficient for a social choice function $f(\cdot)$ to be truthfully implementable in dominant strategies, can be usefully thought of in terms of a certain *weak preference reversal property*. In particular, consider any agent i and any pair of possible types for i , θ'_i and θ''_i . If truth telling is a dominant strategy for agent i , then for any $\theta_{-i} \in \Theta_{-i}$ we must have

$$u_i(f(\theta'_i, \theta_{-i}), \theta'_i) \geq u_i(f(\theta''_i, \theta_{-i}), \theta'_i)$$

and

$$u_i(f(\theta''_i, \theta_{-i}), \theta''_i) \geq u_i(f(\theta'_i, \theta_{-i}), \theta''_i).$$

That is, agent i 's preference ranking of $f(\theta'_i, \theta_{-i})$ and $f(\theta''_i, \theta_{-i})$ must *weakly reverse* when his type changes from θ'_i to θ''_i , with him weakly preferring alternative $f(\theta'_i, \theta_{-i})$ to $f(\theta''_i, \theta_{-i})$ when his type is θ'_i , but weakly preferring alternative $f(\theta''_i, \theta_{-i})$ to $f(\theta'_i, \theta_{-i})$ when his type is θ''_i . In the reverse direction, if this weak preference reversal property holds for all $\theta_{-i} \in \Theta_{-i}$ and all pairs $\theta'_i, \theta''_i \in \Theta_i$, then truth telling is indeed a dominant strategy for agent i (check this in Exercise 23.C.1).

This weak preference reversal property can be succinctly stated using agent i 's *lower contour sets*. Define the lower contour set of alternative x when agent i has type θ_i by (see Section 3.B):

$$L_i(x, \theta_i) = \{z \in X: u_i(x, \theta_i) \geq u_i(z, \theta_i)\}.$$

Using this lower contour set we get the characterization of the set of social choice functions that can be truthfully implemented in dominant strategies that is given in Proposition 23.C.2.

Proposition 23.C.2: The social choice function $f(\cdot)$ is truthfully implementable in dominant strategies if and only if for all i , all $\theta_{-i} \in \Theta_{-i}$, and all pairs of types for agent i , θ'_i and $\theta''_i \in \Theta_i$, we have

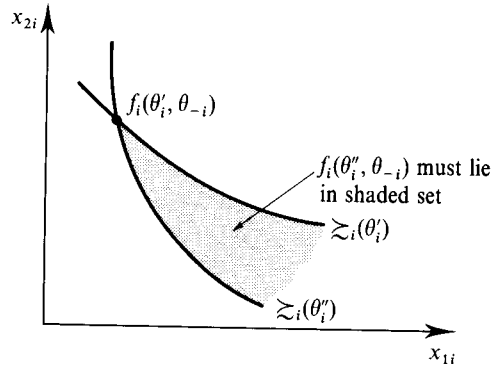
$$f(\theta'_i, \theta_{-i}) \in L_i(f(\theta'_i, \theta_{-i}), \theta'_i) \quad \text{and} \quad f(\theta'_i, \theta_{-i}) \in L_i(f(\theta''_i, \theta_{-i}), \theta''_i). \quad (23.C.6)$$

The idea behind this preference reversal characterization of the social choice functions that can be truthfully implemented in dominant strategies is illustrated in Figures 23.C.1 and 23.C.2. In Figure 23.C.1, we represent the social choice function $f(\cdot)$ for each possible configuration of types (θ_1, θ_2) in a situation in which there are two agents ($I = 2$), two possible values of θ_1 , and three possible values of θ_2 . Consider agent 1's incentives to tell the truth. If truth telling is a weakly dominant strategy for agent 1, then when his type changes from θ'_1 to θ''_1 , he must experience a weak preference reversal between outcomes $f(\theta'_1, \theta_2)$ and $f(\theta''_1, \theta_2)$ for each possible value of θ_2 . A similar point applies for agent 2.

		θ_2		
		θ'_2	θ''_2	θ'''_2
θ_1	θ'_1	$f(\theta'_1, \theta'_2)$	$f(\theta'_1, \theta''_2)$	$f(\theta'_1, \theta'''_2)$
	θ''_1	$f(\theta''_1, \theta'_2)$	$f(\theta''_1, \theta''_2)$	$f(\theta''_1, \theta'''_2)$

Figure 23.C.1

For agent 1 to find truth telling to be his dominant strategy, he must experience a weak preference reversal between outcomes $f(\theta'_1, \theta_2)$ and $f(\theta''_1, \theta_2)$ when his type changes from θ'_1 to θ''_1 , for each possible θ_2 .

**Figure 23.C.2**

The weak preference reversal property of Proposition 23.C.2 when preferences satisfy the single-crossing property.

Figure 23.C.2 depicts a change in some agent i 's type from θ'_i to θ''_i in an exchange setting in which agent i 's preferences satisfy the *single-crossing property* that we discussed in Sections 13.C and 14.C. In the figure, we denote agent i 's allocation in outcome $f(\theta_1, \theta_2)$ by $f_i(\theta_1, \theta_2)$. According to Proposition 23.C.2, $f_i(\theta''_i, \theta_{-i})$ must lie in the shaded region of the figure if truth telling is to be a dominant strategy for agent i . Thus, the characterization in Proposition 23.C.2 can be seen as a multiperson extension of the truth-telling constraints that we encountered in Section 14.C (here they must hold for every possible $\theta_{-i} \in \Theta_{-i}$).

In the remainder of this section we explore in more detail the characteristics of social choice functions that can be truthfully implemented in dominant strategies.

The Gibbard–Satterthwaite Theorem

The Gibbard–Satterthwaite theorem was discovered independently in the early 1970s by the two named authors [Gibbard (1973), Satterthwaite (1975)]. It is an impossibility result similar in spirit to Arrow's theorem (Proposition 21.C.1), and has shaped the course of research on incentives and implementation to a great extent. It shows that for a very general class of problems there is no hope of implementing satisfactory social choice functions in dominant strategies.

In what follows, we let $\mathcal{P}$ denote the set of all rational preference relations $\succsim$ on X having the property that no two alternatives are indifferent, and we recall that $\mathcal{R}_i = \{\succsim_i: \succsim_i = \succsim_i(\theta_i) \text{ for some } \theta_i \in \Theta_i\}$ is agent i 's set of possible ordinal preference relations over X . We denote by $f(\Theta)$ the image of $f(\cdot)$; that is, $f(\Theta) = \{x \in X: f(\theta) = x \text{ for some } \theta \in \Theta\}$. In Definitions 23.C.4 and 23.C.5 we also recall two properties of social choice functions introduced and discussed in Section 21.E.

Definition 23.C.4: The social choice function $f(\cdot)$ is *dictatorial* if there is an agent i such that, for all $\theta = (\theta_1, \dots, \theta_I) \in \Theta$,

$$f(\theta) \in \{x \in X: u_i(x, \theta_i) \geq u_i(y, \theta_i) \text{ for all } y \in X\}.$$

In words: A social choice function is dictatorial if there is an agent i such that $f(\cdot)$ always chooses one of i 's top-ranked alternatives.

Definition 23.C.5: The social choice function $f(\cdot)$ is *monotonic* if, for any θ , if θ' is such that $L_i(f(\theta), \theta_i) \subset L_i(f(\theta'), \theta'_i)$ for all i [i.e., if $L_i(f(\theta), \theta_i)$ is weakly included in $L_i(f(\theta'), \theta'_i)$ for all i], then $f(\theta') = f(\theta)$.

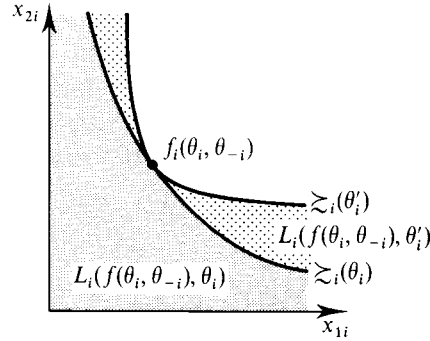


Figure 23.C.3

If $f(\cdot)$ is monotonic, then $f(\theta'_i, \theta_{-i}) = f(\theta_i, \theta_{-i})$.

Monotonicity requires the following: Suppose that $f(\theta) = x$, and that the I agents' types change to a $\theta' = (\theta'_1, \dots, \theta'_I)$ with the property that no agent finds that some alternative that was weakly worse for him than x when his type was θ_i becomes strictly preferred to x when his type is θ'_i . Then x must still be the social choice. The monotonicity property is illustrated in Figure 23.C.3 for an exchange setting. In the figure, $f_i(\theta_i, \theta_{-i})$ represents agent i 's allocation in outcome $f(\theta_i, \theta_{-i})$. The figure depicts a change in agent i 's type from θ_i to a θ'_i having the property that $L_i(f(\theta_i, \theta_{-i}), \theta_i) \subset L_i(f(\theta_i, \theta_{-i}), \theta'_i)$. If $f(\cdot)$ is monotonic, then $f(\theta'_i, \theta_{-i}) = f(\theta_i, \theta_{-i})$.

With these definitions we now state and prove the Gibbard–Satterthwaite theorem.

Proposition 23.C.3: (The Gibbard–Satterthwaite Theorem) Suppose that X is finite and contains at least three elements, that $\mathcal{R}_i = \mathcal{P}$ for all i , and that $f(\Theta) = X$.¹⁹ Then the social choice function $f(\cdot)$ is truthfully implementable in dominant strategies if and only if it is dictatorial.

Proof: It is immediate that a dictatorial $f(\cdot)$ is truthfully implementable (check for yourself that every agent will tell the truth). We now show that if $f(\cdot)$ is truthfully implementable in dominant strategies then it must be dictatorial. The argument consists of three steps.

Step 1: If $\mathcal{R}_i = \mathcal{P}$ for all i , and $f(\cdot)$ is truthfully implementable in dominant strategies, then $f(\cdot)$ is monotonic.

Consider two profiles of types θ and θ' such that $L_i(f(\theta), \theta_i) \subset L_i(f(\theta), \theta'_i)$ for all i . We want to show that $f(\theta') = f(\theta)$. Let us begin by determining $f(\theta'_1, \theta_2, \dots, \theta_I)$. By Proposition 23.C.2 we know that we must have $f(\theta'_1, \theta_2, \dots, \theta_I) \in L_1(f(\theta), \theta_1)$. Hence, $f(\theta'_1, \theta_2, \dots, \theta_I) \in L_1(f(\theta), \theta'_1)$. But Proposition 23.C.2 also implies that $f(\theta) \in L_1(f(\theta'_1, \theta_2, \dots, \theta_I), \theta'_1)$. Since, by hypothesis, no two alternatives can be indifferent in preference relation $\succeq_1(\theta'_1)$, this must imply that $f(\theta'_1, \theta_2, \dots, \theta_I) = f(\theta)$. The same line of argument can be used to show next that $f(\theta'_1, \theta'_2, \theta_3, \dots, \theta_I) = f(\theta)$. Indeed, proceeding iteratively, we establish that $f(\theta') = f(\theta)$. Thus, $f(\cdot)$ is monotonic.

Step 2: If $\mathcal{R}_i = \mathcal{P}$ for all i , $f(\cdot)$ is monotonic, and $f(\Theta) = X$, then $f(\cdot)$ is ex post efficient.

19. Strictly speaking, finiteness of the set X is not required for the result. But in the absence of finiteness, our assumption that agents are expected utility maximizers may not be compatible with the condition that $\mathcal{R}_i = \mathcal{P}$ (e.g., if $X = \mathbb{R}_+^L$, the lexicographic preference relation studied in Example 3.C.1 is a strict preference relation that is not representable by a utility function). For a proof that Proposition 23.C.3 continues to be true if we let X be an arbitrary set and $\mathcal{R}_i$ be the set of all continuous preferences on X , see Barberà and Peleg (1990).

To verify this, suppose not. Then there is a $\theta \in \Theta$ and a $y \in X$ such that $u_i(y, \theta_i) > u_i(f(\theta), \theta_i)$ for all i (recall that no two alternatives can be indifferent). Because $f(\Theta) = X$, there exists a $\theta' \in \Theta$ such that $f(\theta') = y$. Now choose a vector of types $\theta'' \in \Theta$ such that, for all i , $u_i(y, \theta''_i) > u_i(f(\theta), \theta''_i) > u_i(z, \theta''_i)$ for all $z \neq f(\theta), y$. (Remember that all preferences in $\mathcal{P}$ are possible.) Since $L_i(y, \theta''_i) \subset L_i(f(\theta), \theta''_i)$ for all i , monotonicity implies that $f(\theta'') = y$. But, since $L_i(f(\theta), \theta_i) \subset L_i(f(\theta), \theta''_i)$ for all i , monotonicity also implies that $f(\theta'') = f(\theta)$: a contradiction because $y \neq f(\theta)$. Hence, $f(\cdot)$ must be ex post efficient.

Step 3: A social choice function $f(\cdot)$ that is monotonic and ex post efficient is necessarily dictatorial.

Step 3 follows directly from Proposition 21.E.1.

Together, steps 1 to 3 establish the result. ■

It should be noted that the conclusion of Proposition 23.C.3 does *not* follow if X contains two elements. For example, in this case, a majority voting social choice function (see Section 21.E) is both nondictatorial and truthfully implementable in dominant strategies (Exercise 23.C.2).

Note also that when $\mathcal{R}_i = \mathcal{P}$ for all i , any ex post efficient social choice function *must* have $f(\Theta) = X$ (verify this in Exercise 23.C.3). Thus, the Gibbard–Satterthwaite theorem tells us that when $\mathcal{R}_i = \mathcal{P}$ for all i , and X contains more than two elements, the only ex post efficient social choice functions that are truthfully implementable in dominant strategies are dictatorial social choice functions.

Given this negative conclusion, if we are to have any hope of implementing desirable social choice functions, we must either weaken the demands of our implementation concept by accepting implementation by means of less robust equilibrium notions (such as Bayesian Nash equilibria) or we must focus on more restricted environments. In the remainder of this section, we follow the latter course, studying the possibilities for implementing desirable social choice functions in dominant strategies when preferences take a quasilinear form. Section 23.D explores the former possibility: It studies implementation in Bayesian Nash equilibria.

Proposition 23.C.3 is readily extended in two ways. First, the result's conclusion still follows whenever $\mathcal{R}_i$ contains $\mathcal{P}$ (the set of all rational preference relations having the property that no two alternatives are indifferent), and so it extends to environments in which individual indifference is possible. This is stated formally in Corollary 23.C.1.

Corollary 23.C.1: Suppose that X is finite and contains at least three elements, that $\mathcal{P} \subset \mathcal{R}_i$ for all i , and that $f(\Theta) = X$. Then the social choice function $f(\cdot)$ is truthfully implementable in dominant strategies if and only if it is dictatorial.

Proof: It is again immediate that a dictatorial social choice function is truthfully implementable. We now show that under the stated hypotheses $f(\cdot)$ must be dictatorial if it is truthfully implementable.

An implication of Proposition 23.C.3 is that there must be an agent h such that $f(\theta) \in \{x \in X : u_h(x, \theta_h) \geq u_h(y, \theta_h) \text{ for all } y \in X\}$ whenever $\succsim_i(\theta_i) \in \mathcal{P}$ for all i (see Exercise 23.C.4). Without loss of generality, let this be agent I . Suppose now that the result is not true. Then there is a profile of types $\theta' \in \Theta$ such that $f(\theta') \notin \{x \in X : u_I(x, \theta'_I) \geq u_I(y, \theta'_I) \text{ for all } y \in X\}$.

$y \in X\}$. Let $z \in \{x \in X: u_i(x, \theta_i) \geq u_i(y, \theta_i) \text{ for all } y \in X\}$. Now consider a profile of types $\theta'' \in \Theta$ such that (i) $\succsim_i(\theta_i'') \in \mathcal{P}$ for all $i = 1, \dots, I$; (ii) for all agents $i \neq I$, $u_i(f(\theta'), \theta_i'') > u_i(z, \theta_i'')$ and $u_i(z, \theta_i'') > u_i(x, \theta_i'')$ for all $x \notin \{f(\theta'), z\}$; and (iii) $u_I(z, \theta_I'') > u_I(f(\theta'), \theta_I'') > u_I(x, \theta_I'')$ for all $x \notin \{f(\theta'), z\}$. Consider the profile of types $(\theta_1'', \theta_2'', \dots, \theta_I'')$. By Proposition 23.C.2, we must have $f(\theta'') \in L_1(f(\theta_1'', \theta_2'', \dots, \theta_I''), \theta_1'')$, and so it must be that $f(\theta_1'', \theta_2'', \dots, \theta_I'') = f(\theta')$. The same argument can be applied iteratively for all $i \neq I$ to show that $f(\theta_1'', \dots, \theta_{I-1}'', \theta_I'') = f(\theta')$. Next, note that (by Proposition 23.C.2) we must have $f(\theta_1'', \dots, \theta_{I-1}'', \theta_I'') \in L_I(f(\theta''), \theta_I'')$. Hence, $f(\theta'') \in \{z, f(\theta')\}$. But (by Proposition 23.C.2) we must also have $f(\theta'') \in L_I(f(\theta_1'', \dots, \theta_{I-1}'', \theta_I''), \theta_I'')$, and since $u_I(z, \theta_I'') > u_I(f(\theta'), \theta_I'')$ this means we cannot have $f(\theta'') = z$. Hence, $f(\theta'') = f(\theta')$. But, since $u_I(z, \theta_I'') > u_I(f(\theta'), \theta_I'')$, this contradicts agent I being a dictator whenever $\succsim_i(\theta_i) \in \mathcal{P}$ for all i . ■

As our second extension, we can derive a related dictatorship result for social choice functions whose image $f(\Theta)$ is smaller than X . We first offer Definition 23.C.6.

Definition 23.C.6: The social choice function $f(\cdot)$ is *dictatorial on set* $\hat{X} \subset X$ if there exists an agent i such that, for all $\theta = (\theta_1, \dots, \theta_I) \in \Theta$, $f(\theta) \in \{x \in \hat{X}: u_i(x, \theta_i) \geq u_i(y, \theta_i) \text{ for all } y \in \hat{X}\}$.

This weaker notion of dictatorship requires only that $f(\cdot)$ select one of the dictator's most preferred alternatives in $\hat{X}$, rather than in X .

Corollary 23.C.2: Suppose that X is finite, that the number of elements in $f(\Theta)$ is at least three, and that $\mathcal{P} \subset \mathcal{R}_i$ for all $i = 1, \dots, I$. Then $f(\cdot)$ is truthfully implementable in dominant strategies if and only if it is dictatorial on the set $f(\Theta)$.

Proof: It is immediate that $f(\cdot)$ is truthfully implementable if it is dictatorial on the set $f(\Theta)$, and so we now show that under the stated hypotheses $f(\cdot)$ must be dictatorial on set $f(\Theta)$. If $f: \Theta \rightarrow X$ is truthfully implementable in dominant strategies when the set of alternatives is X , then the social choice function $\hat{f}: \Theta \rightarrow f(\Theta)$ which has $\hat{f}(\theta) = f(\theta)$ for all $\theta \in \Theta$ is truthfully implementable in dominant strategies when the set of alternatives is $f(\Theta)$. By Corollary 23.C.1, $\hat{f}(\cdot)$ must be dictatorial. Hence, $f(\cdot)$ is dictatorial on the set $f(\Theta)$. ■

The implication flowing from Corollary 23.C.2 is therefore this: When $\mathcal{R}_i \subset \mathcal{P}$ for all i , the set of social choice functions which have an image that contains at least three elements and which are truthfully implementable in dominant strategies is exactly the set of social choice functions that can be implemented (indirectly) by restricting the set of possible choices to some subset $\hat{X} \subset X$ and assigning a single agent i to choose from within this set.

Quasilinear Environments: Groves–Clarke Mechanisms

In this subsection we focus on the special, but much studied, class of environments in which agents have quasilinear preferences. In particular, an alternative is now a vector $x = (k, t_1, \dots, t_I)$, where k is an element of a finite set K , to be called the “project choice,” and $t_i \in \mathbb{R}$ is a transfer of a numeraire commodity (“money”) to agent i . Agent i 's utility function takes the quasilinear form

$$u_i(x, \theta_i) = v_i(k, \theta_i) + (\bar{m}_i + t_i),$$

where $\bar{m}_i$ is agent i 's endowment of the numeraire. We assume that we are dealing with a closed system in which the I agents have no outside source of financing. The set

of alternatives is therefore²⁰

$$X = \{(k, t_1, \dots, t_I) : k \in K, t_i \in \mathbb{R} \text{ for all } i, \text{ and } \sum_i t_i \leq 0\}.$$

Note that this environment encompasses the cases studied in Examples 23.B.3 and 23.B.4:

Example 23.C.1: A Public Project. We can fit a generalized version of the public project setting of Example 23.B.3 into the framework outlined above. To do so, let K contain the possible levels of a public project (e.g., if $K = \{0, 1\}$, then either the project is “not done” or “done”) and denote by $c(k)$ the cost of project level $k \in K$. Suppose that $\tilde{v}_i(k, \theta_i)$ is agent i 's gross benefit from project level k and that, in the absence of any other transfers, projects will be financed through equal contribution [i.e., each agent i will pay the amount $c(k)/I$].²¹ Then, we can write agent i 's net benefit from project level k when his type is θ_i as $v_i(k, \theta_i) = \tilde{v}_i(k, \theta_i) - (c(k)/I)$. The t_i 's are now transfers over and above the payments $c(k)/I$. ■

Example 23.C.2: Allocation of a Single Unit of an Indivisible Private Good. Consider the environment described in Example 23.B.4 in which an indivisible unit of a private good is to be allocated to one of I agents. Here the “project choice” $k = (y_1, \dots, y_I)$ represents the allocation of the private good and $K = \{(y_1, \dots, y_I) : y_i \in \{0, 1\} \text{ for all } i \text{ and } \sum_i y_i = 1\}$. Agent i 's valuation function takes the form $v_i(k, \theta_i) = \theta_i y_i$. ■

A social choice function in this quasilinear environment takes the form $f(\cdot) = (k(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ where, for all $\theta \in \Theta$, $k(\theta) \in K$ and $\sum_i t_i(\theta) \leq 0$. Note that if the social choice function $f(\cdot)$ is ex post efficient then, for all $\theta \in \Theta$, $k(\theta)$ must satisfy

$$\sum_{i=1}^I v_i(k(\theta), \theta_i) \geq \sum_{i=1}^I v_i(k, \theta_i) \quad \text{for all } k \in K. \quad (23.C.7)$$

We begin with a result that identifies a class of social choice functions that satisfy (23.C.7) and that are truthfully implementable in dominant strategies.

Proposition 23.C.4: Let $k^*(\cdot)$ be a function satisfying (23.C.7). The social choice function $f(\cdot) = (k^*(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ is truthfully implementable in dominant strategies if, for all $i = 1, \dots, I$,

$$t_i(\theta) = \left[\sum_{j \neq i} v_j(k^*(\theta), \theta_j) \right] + h_i(\theta_{-i}), \quad (23.C.8)$$

where $h_i(\cdot)$ is an arbitrary function of θ_{-i} .

Proof: If truth is not a dominant strategy for some agent i , then there exist $\theta_i, \hat{\theta}_i$, and θ_{-i} such that

$$v_i(k^*(\hat{\theta}_i, \theta_{-i}), \theta_i) + t_i(\hat{\theta}_i, \theta_{-i}) > v_i(k^*(\theta_i, \theta_{-i}), \theta_i) + t_i(\theta_i, \theta_{-i}).$$

20. Observe that X is not a compact set. This explains what might appear as a small paradox: in this setting, there are no dictatorial social choice functions because any agent i , when allowed to pick his best alternative in X , faces no bound on how much money he can extract from the other agents.

21. Nothing we do depends on this choice for the “base” method of contribution.

Substituting from (23.C.8) for $t_i(\hat{\theta}_i, \theta_{-i})$ and $t_i(\theta_i, \theta_{-i})$, this implies that

$$\sum_{j=1}^I v_j(k^*(\hat{\theta}_i, \theta_{-i}), \theta_j) > \sum_{j=1}^I v_j(k^*(\theta), \theta_j),$$

which contradicts $k^*(\cdot)$ satisfying (23.C.7). Thus, $f(\cdot)$ must be truthfully implementable in dominant strategies. ■

A direct revelation mechanism $\Gamma = (\Theta_1, \dots, \Theta_I, f(\cdot))$ in which $f(\cdot) = (k(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ satisfies (23.C.7) and (23.C.8) is known as a *Groves mechanism* or *Groves scheme* [after Groves (1973)].²² In a Groves mechanism, given the announcements θ_{-i} of agents $j \neq i$, agent i 's transfer depends on his announced type only through his announcement's effect on the project choice $k^*(\theta)$. Moreover, the change in agent i 's transfer that results when his announcement changes the project decision k is exactly equal to the effect of this change in k on agents $j \neq i$. Put differently, the change in agent i 's transfer reflects exactly the *externality* that he is imposing on the other agents. As a result, agent i is led to internalize this externality and make an announcement—namely, truth—that leads to a level of k that maximizes the I agents' joint payoff from the project, $\sum_i v_i(k, \theta_i)$.

A special case of a Groves mechanism was discovered independently by Clarke (1971) and is known as the *Clarke*, or *pivotal*, *mechanism*. This mechanism corresponds to the case in which $h_i(\theta_{-i}) = -\sum_{j \neq i} v_j(k_{-i}^*(\theta_{-i}), \theta_j)$ where, for all $\theta_{-i} \in \Theta_{-i}$, $k_{-i}^*(\theta_{-i})$ satisfies

$$\sum_{j \neq i} v_j(k_{-i}^*(\theta_{-i}), \theta_j) \geq \sum_{j \neq i} v_j(k, \theta_j) \quad \text{for all } k \in K.$$

That is, $k_{-i}^*(\theta_{-i})$ is the project level that would be ex post efficient if there were only the $I - 1$ agents $j \neq i$. Agent i 's transfer in the Clarke mechanism is then given by

$$t_i(\theta) = \left[\sum_{j \neq i} v_j(k^*(\theta), \theta_j) \right] - \left[\sum_{j \neq i} v_j(k_{-i}^*(\theta_{-i}), \theta_j) \right].$$

Note that agent i 's transfer is 0 if his announcement does not change the project decision relative to what would be ex post efficient for agents $j \neq i$ in isolation [i.e., if $k^*(\theta) = k_{-i}^*(\theta_{-i})$], and is negative if it does change the project decision [i.e., if $k^*(\theta) \neq k_{-i}^*(\theta_{-i})$], that is, if agent i is "pivotal" to the efficient project choice. Thus, in the Clarke mechanism agent i pays a tax equal to his effect on the other agents if he is pivotal to the project decision, and he pays nothing otherwise.²³

22. We will sometimes be a little loose in our terminology and simply refer to a social choice function $f(\cdot)$ satisfying (23.C.7) and (23.C.8) as a Groves mechanism.

23. Note that the social choice function of the Clarke mechanism satisfies the feasibility condition that $\sum_i t_i(\theta) \leq 0$ for all θ . Indeed, examining (23.C.8), we see that a sufficient (but not necessary) condition for a Groves scheme to satisfy the condition that $\sum_i t_i(\theta) \leq 0$ for all θ is that

$$h_i(\theta_{-i}) \leq -\sum_{j \neq i} v_j(k_{-i}^*(\theta_{-i}), \theta_j) \quad \text{for all } \theta_{-i} \in \Theta_{-i}.$$

It is interesting to note that in the case of allocation of a single indivisible unit of a private good, the Clarke mechanism is precisely the social choice function implemented by the second-price sealed-bid auction (see Example 23.B.6). In particular: (i) $k^*(\theta)$ is the allocation rule that gives the item to the agent with the highest valuation; (ii) agent i is pivotal precisely when he is the buyer with the highest valuation; and (iii) when he is pivotal his “tax” is exactly equal to the second-highest valuation (in particular, in this case $\sum_{j \neq i} v_j(k^*(\theta), \theta_j) = 0$, and $\sum_{j \neq i} v_j(k^*(\theta_{-i}), \theta_j)$ is equal to the amount of the second-highest valuation).

We have seen so far that social choice functions satisfying (23.C.7) and (23.C.8) are truthfully implementable in dominant strategies. Are these the only social choice functions satisfying (23.C.7) that are truthfully implementable? The result given in Proposition 23.C.5 [due to Green and Laffont (1979)] provides one set of conditions under which the answer is “yes.”²⁴ In it, we let $\mathcal{V}$ denote the set of all possible functions $v: K \rightarrow \mathbb{R}$.

Proposition 23.C.5: Suppose that for each agent $i = 1, \dots, I$, $\{v_i(\cdot, \theta_i): \theta_i \in \Theta_i\} = \mathcal{V}$; that is, every possible valuation function from K to $\mathbb{R}$ arises for some $\theta_i \in \Theta_i$. Then a social choice function $f(\cdot) = (k^*(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ in which $k^*(\cdot)$ satisfies (23.C.7) is truthfully implementable in dominant strategies only if $t_i(\cdot)$ satisfies (23.C.8) for all $i = 1, \dots, I$.

Proof: Note first that we can always write

$$t_i(\theta_i, \theta_{-i}) = \sum_{j \neq i} v_j(k^*(\theta_i, \theta_{-i}), \theta_j) + h_i(\theta_i, \theta_{-i}). \quad (23.C.9)$$

What we want to show, then, is that the function $h_i(\cdot)$ must in fact be independent of θ_i if $f(\cdot)$ is truthfully implementable in dominant strategies. Suppose that it is not; that is, that $f(\cdot)$ is truthfully implementable in dominant strategies but that for some $\theta_i, \hat{\theta}_i$, and θ_{-i} , we have $h_i(\theta_i, \theta_{-i}) \neq h_i(\hat{\theta}_i, \theta_{-i})$. We now consider two distinct cases.

(i) $k^*(\theta_i, \theta_{-i}) = k^*(\hat{\theta}_i, \theta_{-i})$: If $f(\cdot)$ is truthfully implementable in dominant strategies, then by (23.C.3) we have

$$v_i(k^*(\theta_i, \theta_{-i}), \theta_i) + t_i(\theta_i, \theta_{-i}) \geq v_i(k^*(\hat{\theta}_i, \theta_{-i}), \theta_i) + t_i(\hat{\theta}_i, \theta_{-i})$$

and

$$v_i(k^*(\hat{\theta}_i, \theta_{-i}), \hat{\theta}_i) + t_i(\hat{\theta}_i, \theta_{-i}) \geq v_i(k^*(\theta_i, \theta_{-i}), \hat{\theta}_i) + t_i(\theta_i, \theta_{-i}).$$

Since, $k^*(\theta_i, \theta_{-i}) = k^*(\hat{\theta}_i, \theta_{-i})$, these two inequalities imply that $t_i(\theta_i, \theta_{-i}) = t_i(\hat{\theta}_i, \theta_{-i})$, and so by (23.C.9) we have $h_i(\theta_i, \theta_{-i}) = h_i(\hat{\theta}_i, \theta_{-i})$: a contradiction.

(ii) $k^*(\theta_i, \theta_{-i}) \neq k^*(\hat{\theta}_i, \theta_{-i})$: Suppose, without loss of generality, that $h_i(\theta_i, \theta_{-i}) > h_i(\hat{\theta}_i, \theta_{-i})$. Consider the type $\theta_i^e \in \Theta_i$ such that

$$v_i(k, \theta_i^e) = \begin{cases} -\sum_{j \neq i} v_j(k^*(\theta_i, \theta_{-i}), \theta_j) & \text{if } k = k^*(\theta_i, \theta_{-i}) \\ -\sum_{j \neq i} v_j(k^*(\hat{\theta}_i, \theta_{-i}), \theta_j) + \varepsilon & \text{if } k = k^*(\hat{\theta}_i, \theta_{-i}) \\ -\infty & \text{otherwise.} \end{cases} \quad (23.C.10)$$

24. For another, see the small-type discussion at the end of this section and Exercise 23.C.10.

We will argue that for a sufficiently small $\varepsilon > 0$, type θ_i^ε will strictly prefer to falsely report that he is type θ_i when the other agents' types are θ_{-i} . To see this, note first that $k^*(\theta_i^\varepsilon, \theta_{-i}) = k^*(\hat{\theta}_i, \theta_{-i})$ since setting $k = k^*(\hat{\theta}_i, \theta_{-i})$ maximizes $v_i(k, \theta_i^\varepsilon) + \sum_{j \neq i} v_j(k, \theta_j)$. Thus, truth telling being a dominant strategy requires that

$$v_i(k^*(\hat{\theta}_i, \theta_{-i}), \theta_i^\varepsilon) + t_i(\theta_i^\varepsilon, \theta_{-i}) \geq v_i(k^*(\theta_i, \theta_{-i}), \theta_i) + t_i(\theta_i, \theta_{-i}),$$

or, substituting, from (23.C.9) and (23.C.10),

$$\varepsilon + h_i(\theta_i^\varepsilon, \theta_{-i}) \geq h_i(\theta_i, \theta_{-i}).$$

But by the logic of part (i), $h_i(\theta_i^\varepsilon, \theta_{-i}) = h_i(\hat{\theta}_i, \theta_{-i})$ because $k^*(\theta_i^\varepsilon, \theta_{-i}) = k^*(\hat{\theta}_i, \theta_{-i})$. This gives

$$\varepsilon + h_i(\hat{\theta}_i, \theta_{-i}) \geq h_i(\theta_i, \theta_{-i}). \quad (23.C.11)$$

By hypothesis we have $h(\theta_i, \theta_{-i}) > h(\hat{\theta}_i, \theta_{-i})$, and so (23.C.11) must be violated for small enough $\varepsilon > 0$. This completes the proof. ■

Thus, when all possible functions $v_i(\cdot)$ can arise for some $\theta_i \in \Theta_i$, the only social choice functions satisfying (23.C.7) that are truthfully implementable in dominant strategies are those in the Groves class.

Groves mechanisms and budget balance

Up to this point, we have studied whether we can implement in dominant strategies a social choice function that always results in an efficient choice of k [one satisfying (23.C.7)]. But ex post efficiency also requires that none of the numeraire be wasted, that is, that we satisfy the *budget balance condition*:

$$\sum_i t_i(\theta) = 0 \quad \text{for all } \theta \in \Theta. \quad (23.C.12)$$

We now briefly explore when fully ex post efficient social choice functions [those satisfying both (23.C.7) and (23.C.12)] can be truthfully implemented in dominant strategies.

Unfortunately, in many cases it is impossible to truthfully implement fully ex post efficient social choice functions in dominant strategies. For example, the result [due to Green and Laffont (1979)] in Proposition 23.C.6, whose proof we omit, shows that if the set of possible types for each agent is sufficiently rich, then no social choice functions that are truthfully implementable in dominant strategies are ex post efficient.²⁵

Proposition 23.C.6: Suppose that for each agent $i = 1, \dots, I$, $\{v_i(\cdot, \theta_i) : \theta_i \in \Theta_i\} = \mathcal{V}$; that is, every possible valuation function from K to $\mathbb{R}$ arises for some $\theta_i \in \Theta_i$. Then there is no social choice function $f(\cdot) = (k^*(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ that is truthfully implementable in dominant strategies and is ex post efficient, that is, that satisfies (23.C.7) and (23.C.12).

Thus, under the hypotheses of Proposition 23.C.6, the presence of private information means that the I agents must either accept some waste of the numeraire

25. For another negative result, see the small-type discussion at the end of this section and Exercise 23.C.10.

[i.e., have $\sum_i t_i(\theta) < 0$ for some θ , as in the Clarke mechanism] or give up on always having an efficient project selection [i.e., have a project selection $k(\theta)$ that does not satisfy (23.C.7) for some θ].

One special case in which a more positive result does obtain arises when there is at least one agent whose preferences are known. For notational purposes, let this agent be denoted “agent 0”, and let there still be I agents, denoted $i = 1, \dots, I$, whose preferences are private information (so that we are now letting there be $I + 1$ agents in total). The simplest case of this phenomenon, of course, occurs when agent 0 has no preferences over the project choice k , that is, when his preferences are $u_i(x) = \bar{m}_0 + t_0$. We saw one example of this kind in Example 23.B.4 when we considered auction settings (agent 0 is then the seller). Another example arises in the case of a public project when the project affects only a subset of the agents in the economy (so that agent 0 represents all of the other agents in the economy).

When there is such an agent, ex post efficiency of the social choice function still requires that (23.C.7) be satisfied; but now ex post efficiency is compatible with *any* transfer functions $t_1(\cdot), \dots, t_I(\cdot)$ for the I agents with private information, as long as we set $t_0(\theta) = -\sum_{i \neq 0} t_i(\theta)$ for all θ . That is, in this $(I + 1)$ -agent setting, the Groves mechanisms identified in Proposition 23.C.4 (in which only agents $i = 1, \dots, I$ announce their types) are ex post efficient as long as we set the transfer of agent 0 to be $t_0(\theta) = -\sum_{i \neq 0} t_i(\theta)$ for all θ . In essence, the presence of an “outside” agent 0 who has no private information allows us to break the budget balance condition for those agents who do have privately observed types.

We should offer, however, one immediate caveat to this seemingly positive result: Up to this point, we have not worried about whether agents will find it in their interest to participate in the mechanism. As we will see in Section 23.E, when participation is voluntary, it may be that no ex post efficient social choice function is implementable in dominant strategies even when such an outside agent exists.

The differentiable case

It is common in applications to encounter cases in which $K = \mathbb{R}$, the $v_i(\cdot, \theta_i)$ functions are assumed to be twice continuously differentiable with $\partial^2 v_i(k, \theta_i) / \partial k^2 < 0$ and $\partial^2 v_i(k, \theta_i) / \partial k \partial \theta_i \neq 0$ at all (k, θ_i) , and each θ_i is drawn from an interval $[\underline{\theta}_i, \bar{\theta}_i] \subset \mathbb{R}$ with $\underline{\theta}_i \neq \bar{\theta}_i$. In this case a great deal can be said about the set of social choice functions that can be truthfully implemented in dominant strategies. Exercise 23.C.9 develops this point fully. Here we will simply show how, in this environment, we can easily derive a number of our previous results.

Note first that for any continuously differentiable social choice function $f(\cdot) = (k(\cdot), t_1(\cdot), \dots, t_I(\cdot))$, if truth telling is a dominant strategy for agent i , then agent i 's first-order condition implies that, for all θ_{-i} ,

$$\frac{\partial v_i(k(\theta_i, \theta_{-i}), \theta_i)}{\partial k} \frac{\partial k(\theta_i, \theta_{-i})}{\partial \theta_i} + \frac{\partial t_i(\theta_i, \theta_{-i})}{\partial \theta_i} = 0 \quad (23.C.13)$$

at all $\theta_i \in (\underline{\theta}_i, \bar{\theta}_i)$. Integrating (23.C.13) with respect to the variable θ_i , this implies that for all profiles of types (θ_i, θ_{-i}) we have

$$t_i(\theta_i, \theta_{-i}) = t_i(\underline{\theta}_i, \theta_{-i}) - \int_{\underline{\theta}_i}^{\theta_i} \frac{\partial v_i(k(s, \theta_{-i}), s)}{\partial k} \frac{\partial k(s, \theta_{-i})}{\partial s} ds. \quad (23.C.14)$$

Consider now any social choice function $f(\cdot) = (k^*(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ that satisfies

(23.C.7). Under our present assumptions $k^*(\cdot)$ must satisfy, for all θ ,

$$\sum_{j=1}^I \frac{\partial v_j(k^*(\theta), \theta_j)}{\partial k} = 0. \quad (23.C.15)$$

Moreover, using the implicit function theorem and our assumptions on the $v_i(\cdot)$ functions, we see that $k^*(\cdot)$ is continuously differentiable and that it has nonzero partial derivatives, $\partial k^*(\theta)/\partial \theta_i \neq 0$ for all i .

We now substitute for $\partial v_i(k^*(s, \theta_{-i}), s)/\partial k$ in (23.C.14) using (23.C.15). Doing so, we derive that, for all profiles (θ_i, θ_{-i}) ,

$$\begin{aligned} t_i(\theta_i, \theta_{-i}) &= t_i(\underline{\theta}_i, \theta_{-i}) + \int_{\underline{\theta}_i}^{\theta_i} \left(\sum_{j \neq i} \frac{\partial v_j(k^*(s, \theta_{-i}), \theta_j)}{\partial k} \right) \frac{\partial k(s, \theta_{-i})}{\partial s} ds \\ &= t_i(\underline{\theta}_i, \theta_{-i}) + \int_{k^*(\underline{\theta}_i, \theta_{-i})}^{k^*(\theta_i, \theta_{-i})} \left(\sum_{j \neq i} \frac{\partial v_j(k, \theta_j)}{\partial k} \right) dk \\ &= t_i(\underline{\theta}_i, \theta_{-i}) + \sum_{j \neq i} v_j(k^*(\theta_i, \theta_{-i}), \theta_j) - \sum_{j \neq i} v_j(k^*(\underline{\theta}_i, \theta_{-i}), \theta_j). \end{aligned}$$

But this is true if and only if $t_i(\theta)$ satisfies (23.C.8). Thus, in this setting, Groves mechanisms are the only social choice functions satisfying (23.C.7) that are truthfully implementable in dominant strategies.²⁶

Consider now the question of budget balance when there is no outside agent. We will show that satisfying (23.C.15) and budget balance is impossible in this differentiable setting when $I = 2$ [for $I > 2$ see Laffont and Maskin (1980) and Exercise 23.C.10]. By (23.C.13), for all $\theta = (\theta_1, \theta_2)$, we have

$$\frac{\partial t_1(\theta)}{\partial \theta_1} = - \frac{\partial v_1(k^*(\theta), \theta_1)}{\partial k} \frac{\partial k^*(\theta)}{\partial \theta_1}$$

and

$$\frac{\partial t_2(\theta)}{\partial \theta_2} = - \frac{\partial v_2(k^*(\theta), \theta_2)}{\partial k} \frac{\partial k^*(\theta)}{\partial \theta_2}.$$

Thus, for all $\theta = (\theta_1, \theta_2)$,

$$- \frac{\partial^2 t_1(\theta)}{\partial \theta_1 \partial \theta_2} = \frac{\partial^2 v_1(k^*(\theta), \theta_1)}{\partial k^2} \frac{\partial k^*(\theta)}{\partial \theta_2} \frac{\partial k^*(\theta)}{\partial \theta_1} + \frac{\partial v_1(k^*(\theta), \theta_1)}{\partial k} \frac{\partial^2 k^*(\theta)}{\partial \theta_1 \partial \theta_2} \quad (23.C.16)$$

and

$$- \frac{\partial^2 t_2(\theta)}{\partial \theta_1 \partial \theta_2} = \frac{\partial^2 v_2(k^*(\theta), \theta_2)}{\partial k^2} \frac{\partial k^*(\theta)}{\partial \theta_1} \frac{\partial k^*(\theta)}{\partial \theta_2} + \frac{\partial v_2(k^*(\theta), \theta_2)}{\partial k} \frac{\partial^2 k^*(\theta)}{\partial \theta_1 \partial \theta_2}, \quad (23.C.17)$$

If we have budget balance, then $t_1(\theta) = -t_2(\theta)$ for all θ , and so we must have $\partial^2 t_1(\theta)/\partial \theta_1 \partial \theta_2 = -\partial^2 t_2(\theta)/\partial \theta_1 \partial \theta_2$. But this would imply, by adding (23.C.16) and (23.C.17), and using (23.C.15), that

$$\left[\frac{\partial^2 v_1(k^*(\theta), \theta_1)}{\partial k^2} + \frac{\partial^2 v_2(k^*(\theta), \theta_2)}{\partial k^2} \right] \frac{\partial k^*(\theta)}{\partial \theta_1} \frac{\partial k^*(\theta)}{\partial \theta_2} = 0,$$

which is impossible under our assumptions.

26. This argument generalizes to any case in which $k^*(\cdot)$ is continuously differentiable.

23.D Bayesian Implementation

In this section, we study implementation in *Bayesian Nash equilibrium*.²⁷ Throughout we follow the notation introduced in Section 23.B: The vector of agents' types $\theta = (\theta_1, \dots, \theta_I)$ is drawn from set $\Theta = \Theta_1 \times \dots \times \Theta_I$ according to probability density $\phi(\cdot)$, and agent i 's Bernoulli utility function over the alternatives in X given his type θ_i is $u_i(x, \theta_i)$. We also adopt the notational convention of writing $\theta_{-i} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_I)$ and $\theta = (\theta_i, \theta_{-i})$. A mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ is a collection of I sets $S_1, \dots, S_I$, each S_i containing agent i 's possible actions (or plans of action), and an outcome function $g: S \rightarrow X$, where $S = S_1 \times \dots \times S_I$. As discussed in Section 23.B, the mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ combined with possible types $(\Theta_1, \dots, \Theta_I)$, density $\phi(\cdot)$, and Bernoulli utility functions $(u_1(\cdot), \dots, u_I(\cdot))$ defines a Bayesian game of incomplete information (see Section 8.E). We will also often write $s_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_I)$, $s = (s_i, s_{-i})$, and $s(\cdot) = (s_i(\cdot), s_{-i}(\cdot))$ where $s_{-i}(\cdot) = (s_1(\cdot), \dots, s_{i-1}(\cdot), s_{i+1}(\cdot), \dots, s_I(\cdot))$.

We begin by defining the concept of a Bayesian Nash equilibrium (see also Section 8.E) and specializing Definition 23.B.4 to the notion of implementation in Bayesian Nash equilibrium.²⁸

Definition 23.D.1: The strategy profile $s^*(\cdot) = (s_1^*(\cdot), \dots, s_I^*(\cdot))$ is a *Bayesian Nash equilibrium* of mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ if, for all i and all $\theta_i \in \Theta_i$,

$$E_{\theta_{-i}}[u_i(g(s_i^*(\theta_i), s_{-i}^*(\theta_{-i})), \theta_i) | \theta_i] \geq E_{\theta_{-i}}[u_i(g(\hat{s}_i, s_{-i}^*(\theta_{-i})), \theta_i) | \theta_i]$$

for all $\hat{s}_i \in S_i$.

Definition 23.D.2: The mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ *implements the social choice function* $f(\cdot)$ *in Bayesian Nash equilibrium* if there is a Bayesian Nash equilibrium of Γ , $s^*(\cdot) = (s_1^*(\cdot), \dots, s_I^*(\cdot))$, such that $g(s^*(\theta)) = f(\theta)$ for all $\theta \in \Theta$.

As with implementation in dominant strategies (see Section 23.C), we will see that a social choice function is Bayesian implementable if and only if it is truthfully implementable in the sense given in Definition 23.D.3.

Definition 23.D.3: The social choice function $f(\cdot)$ is *truthfully implementable in Bayesian Nash equilibrium* (or *Bayesian incentive compatible*) if $s_i^*(\theta_i) = \theta_i$ for all $\theta_i \in \Theta_i$ and $i = 1, \dots, I$ is a Bayesian Nash equilibrium of the direct revelation mechanism $\Gamma = (\Theta_1, \dots, \Theta_I, f(\cdot))$. That is, if for all $i = 1, \dots, I$ and all $\theta_i \in \Theta_i$,

$$E_{\theta_{-i}}[u_i(f(\theta_i, \theta_{-i}), \theta_i) | \theta_i] \geq E_{\theta_{-i}}[u_i(f(\hat{\theta}_i, \theta_{-i}), \theta_i) | \theta_i] \quad (23.D.1)$$

for all $\hat{\theta}_i \in \Theta_i$.

The ability to restrict our inquiry, without loss of generality, to the question of whether $f(\cdot)$ is truthfully implementable is a consequence of the *revelation principle for Bayesian Nash equilibrium*.

27. Good sources for further reading on the subject of this section are Myerson (1991) and Fudenberg and Tirole (1991).

28. As in Section 8.E, we restrict our attention to pure strategy equilibria.

Proposition 23.D.1: (*The Revelation Principle for Bayesian Nash Equilibrium*) Suppose that there exists a mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ that implements the social choice function $f(\cdot)$ in Bayesian Nash equilibrium. Then $f(\cdot)$ is truthfully implementable in Bayesian Nash equilibrium.

Proof: If $\Gamma = (S_1, \dots, S_I, g(\cdot))$ implements $f(\cdot)$ in Bayesian Nash equilibrium, then there exists a profile of strategies $s^*(\cdot) = (s_1^*(\cdot), \dots, s_I^*(\cdot))$ such that $g(s^*(\theta)) = f(\theta)$ for all θ , and for all i and all $\theta_i \in \Theta_i$,

$$E_{\theta_{-i}}[u_i(g(s_i^*(\theta_i), s_{-i}^*(\theta_{-i})), \theta_i) | \theta_i] \geq E_{\theta_{-i}}[u_i(g(\hat{s}_i, s_{-i}^*(\theta_{-i})), \theta_i) | \theta_i] \quad (23.D.2)$$

for all $\hat{s}_i \in S_i$. Condition (23.D.2) implies, in particular, that for all i and all $\theta_i \in \Theta_i$,

$$E_{\theta_{-i}}[u_i(g(s_i^*(\theta_i), s_{-i}^*(\theta_{-i})), \theta_i) | \theta_i] \geq E_{\theta_{-i}}[u_i(g(s_i^*(\hat{\theta}_i), s_{-i}^*(\theta_{-i})), \theta_i) | \theta_i] \quad (23.D.3)$$

for all $\hat{\theta}_i \in \Theta_i$. Since $g(s^*(\theta)) = f(\theta)$ for all θ , (23.D.3) means that, for all i and all $\theta_i \in \Theta_i$,

$$E_{\theta_{-i}}[u_i(f(\theta_i, \theta_{-i}), \theta_i) | \theta_i] \geq E_{\theta_{-i}}[u_i(f(\hat{\theta}_i, \theta_{-i}), \theta_i) | \theta_i] \quad (23.D.4)$$

for all $\hat{\theta}_i \in \Theta_i$. But, this is precisely condition (23.D.1), the condition for $f(\cdot)$ to be truthfully implementable in Bayesian Nash equilibrium. ■

The basic idea behind the revelation principle for Bayesian Nash equilibrium parallels that given for the revelation principle for dominant strategy implementation (Proposition 23.C.1): If in mechanism $\Gamma = (S_1, \dots, S_I, g(\cdot))$ each agent finds that, when his type is θ_i , choosing $s_i^*(\theta_i)$ is his best response to the other agents' strategies, then if we introduce a mediator who says "Tell me your type, θ_i , and I will play $s_i^*(\theta_i)$ for you," each agent will find truth telling to be an optimal strategy given that all other agents tell the truth. That is, truth telling will be a Bayesian Nash equilibrium of this direct revelation game.

The implication of the revelation principle is, once again, that to identify the set of implementable social choice functions (now in Bayesian Nash equilibrium) we need only identify those that are truthfully implementable.²⁹

We can note immediately that the Bayesian implementation concept is a strictly weaker notion than the notion of dominant strategy implementation. Since every dominant strategy equilibrium is necessarily a Bayesian Nash equilibrium, any social choice function that is implementable in dominant strategies is a fortiori implementable in Bayesian Nash equilibrium. Intuitively put, when we compare the requirements for truthful implementation of a social choice function $f(\cdot)$ in dominant strategies and in Bayesian Nash equilibrium given in equations (23.C.3) and (23.D.1), respectively, we see that, with Bayesian implementation, truth telling need only give agent i his highest payoff *averaging over all possible types θ_{-i} that might arise for the other agents*. In contrast, the dominant strategy concept requires that truth telling be agent i 's best strategy *for every possible θ_{-i}* . Thus, we can reasonably hope to be

29. Note that Proposition 23.D.1 is what we implicitly relied on in Section 14.C when, in studying the principal-agent problem with hidden information, we restricted our focus to direct revelation mechanisms that induced truth telling by the agent. Formally, Proposition 23.D.1 tells us that the equilibrium outcome arising from any contract between the principal and the agent can be replicated using a direct revelation mechanism that induces the agent to truthfully reveal his type.

able to successfully implement a wider range of social choice functions in Bayesian Nash equilibrium than in dominant strategies. The drawback, of course, is that we can be less confident about this implementation relative to implementation in dominant strategies because it depends on the agents (and any outside mechanism designer) knowing the density $\phi(\cdot)$ of agents' types, as well as on the plausibility of the Nash assumption that the agents' have mutually correct expectations about each others' strategy choices (see Section 8.D).

In what follows in the remainder of this section, we first provide an example that illustrates that we can indeed implement a wider range of social choice functions with Bayesian implementation. Specifically, we show that in the quasilinear environment studied in Section 23.C, whenever the agents' types are statistically independent of one another, we can *always* Bayesian implement at least one ex post efficient social choice function (i.e., one that both has an efficient project choice and satisfies budget balance). In Section 23.C we saw that it may not be possible to accomplish this with dominant strategy implementation (recall Proposition 23.C.6 and the small-type discussion at the end of that section). After showing this, we then examine the properties of Bayesian implementable social choice functions in greater detail, concentrating on the special case in which agents have quasilinear utility functions that are linear in their type. As an application of this analysis, we prove the *revenue equivalence theorem* for auctions.

The Expected Externality Mechanism

Let us return to the quasilinear setting studied in Section 23.C. In particular, an alternative is now a vector $x = (k, t_1, \dots, t_I)$, where k is an element of a finite set K , and $t_i \in \mathbb{R}$ is a transfer of a numeraire commodity ("money") to agent i . Agent i 's utility function takes the quasilinear form

$$u_i(x, \theta_i) = v_i(k, \theta_i) + (\bar{m}_i + t_i), \quad (23.D.5)$$

where $\bar{m}_i$ is agent i 's endowment of the numeraire.³⁰ For simplicity, we shall henceforth normalize $\bar{m}_i = 0$ for all i . We assume here that the I agents have no outside source of financing, and so $X = \{(k, t_1, \dots, t_I) : k \in K, t_i \in \mathbb{R} \text{ for all } i, \text{ and } \sum_i t_i \leq 0\}$. A social choice function in this environment takes the form $f(\cdot) = (k(\cdot), t_1(\cdot), \dots, t_I(\cdot))$. Note that if $f(\cdot)$ is ex post efficient then, for all $\theta \in \Theta$,

$$\sum_i v_i(k(\theta), \theta_i) \geq \sum_i v_i(k, \theta_i) \quad \text{for all } k \in K \quad (23.D.6)$$

and

$$\sum_i t_i(\theta) = 0. \quad (23.D.7)$$

In Proposition 23.C.6 we saw that conditions exist in which no social choice

30. Unlike the analysis in Section 23.C (see Exercise 23.C.11), the developments that follow depend not only on preferences over certain outcomes having a quasilinear form but also on the fact that, with this Bernoulli utility function, each agent i is risk neutral with respect to lotteries over his monetary transfer.

function $f(\cdot) = (k(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ satisfying both (23.D.6) and (23.D.7) is truthfully implementable in dominant strategies. We will now show that it is possible to implement such a social choice function in Bayesian Nash equilibrium whenever the agents' types are statistically independent of one another [i.e., when the density $\phi(\cdot)$ has the form $\phi(\theta) = \phi_1(\theta_1) \times \dots \times \phi_I(\theta_I)$].³¹

To verify this, let $k^*(\cdot)$ satisfy (23.D.6) and consider a social choice function $f(\cdot) = (k^*(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ in which, for all $i = 1, \dots, I$,

$$t_i(\theta) = E_{\hat{\theta}_{-i}} \left[\sum_{j \neq i} v_j(k^*(\theta_i, \hat{\theta}_{-i}, \hat{\theta}_j)) \right] + h_i(\theta_{-i}), \quad (23.D.8)$$

where, for now, we take $h_i(\cdot)$ to be an arbitrary function of θ_{-i} . Note that the expectational term in (23.D.8) represents the *expected benefits* of agents $j \neq i$ when agent i announces his type to be θ_i and agents $j \neq i$ tell the truth. (As such, it is a function of only agent i 's actual announcement θ_i —it is *not* a function of the actual announcements θ_{-i} of agents $j \neq i$.) Thus, the change in agent i 's transfer when he changes his announced type is exactly equal to the *expected externality* of this change on agents $j \neq i$.

We check first that any social choice function $f(\cdot)$ with the form (23.D.8) is Bayesian incentive compatible. To see this, note that when agents $j \neq i$ announce their types truthfully, agent i finds that truth telling is his optimal strategy because (using statistical independence of θ_i and θ_{-i})

$$\begin{aligned} E_{\theta_{-i}}[v_i(k^*(\theta), \theta_i) + t_i(\theta)|\theta_i] &= E_{\theta_{-i}} \left[\sum_{j=1}^I v_j(k^*(\theta), \theta_j) \right] + E_{\theta_{-i}}[h_i(\theta_{-i})] \\ &\geq E_{\theta_{-i}} \left[\sum_{j=1}^I v_j(k^*(\hat{\theta}_i, \theta_{-i}, \theta_j)) \right] + E_{\theta_{-i}}[h_i(\theta_{-i})] \\ &= E_{\theta_{-i}}[v_i(k^*(\hat{\theta}_i, \theta_{-i}, \theta_j) + t_i(\hat{\theta}_i, \theta_{-i})|\theta_i] \end{aligned}$$

for all $\hat{\theta}_i \in \Theta_i$, where the inequality follows because $k^*(\cdot)$ satisfies (23.D.6).

What remains is to show that we can choose the $h_i(\cdot)$ functions (for $i = 1, \dots, I$) so that we also satisfy the budget balance condition (23.D.7). For notational ease, define $\xi_i(\theta_i) = E_{\hat{\theta}_{-i}}[\sum_{j \neq i} v_j(k^*(\theta_i, \hat{\theta}_{-i}, \hat{\theta}_j))]$. We now let

$$h_i(\theta_{-i}) = -\left(\frac{1}{I-1}\right) \sum_{j \neq i} \xi_j(\theta_j), \quad (23.D.9)$$

for $i = 1, \dots, I$. With this choice for the $h_i(\cdot)$ functions, we have

$$\begin{aligned} \sum_i t_i(\theta) &= \sum_i \xi_i(\theta_i) + \sum_i h_i(\theta_{-i}) \\ &= \sum_i \xi_i(\theta_i) - \left(\frac{1}{I-1}\right) \sum_i \sum_{j \neq i} \xi_j(\theta_j) \\ &= \sum_i \xi_i(\theta_i) - \left(\frac{1}{I-1}\right) \sum_i (I-1) \xi_i(\theta_i) \\ &= 0. \end{aligned}$$

31. See Fudenberg and Tirole (1991) for a discussion of the case of correlated types and for further references.

Intuitively, the form of the $h_i(\cdot)$ functions in (23.D.9) can be thought of as follows: We have seen that when the agents' types are $(\theta_1, \dots, \theta_I)$, each agent $i = 1, \dots, I$ receives a payment equal to $\xi_i(\theta_i)$ [the first term in (23.D.8)]. Now, if each agent contributes an equal $1/(I - 1)$ share of all of the other agents' payments, the payments from a given agent i to each of the other $I - 1$ agents will total $[1/(I - 1)] \sum_{j \neq i} \xi_j(\theta_j)$, and agent i will receive from these agents in return payments that total to $\xi_i(\theta_i)$. Agent i 's net transfer will therefore be $\xi_i(\theta_i) - (1/(I - 1)) \sum_{j \neq i} \xi_j(\theta_j)$.

This direct revelation mechanism is known as the *expected externality mechanism* [due to d'Aspremont and Gérard-Varet (1979) and Arrow (1979)]. In summary, we have shown that when agents' Bernoulli utility functions take the form in (23.D.5), and agents' types are statistically independent, there is an ex post efficient social choice function that is implementable in Bayesian Nash equilibrium.

Although this is an interesting result, it is not the end of the story, even when we restrict our attention to Bernoulli utility functions of the form (23.D.5) and a statistically independent distribution of types. The reason is that while the expected externality mechanism implements an ex post efficient social choice function, its transfer functions imply a particular distribution of utility across the various types of the agents, and we may wish to consider other mechanisms, possibly ones involving social choice functions that are not ex post efficient, that alter this distribution.

One reason why this may be important is that, in many applications of interest, agents are free to opt out of the mechanism, and so any mechanism that we wish to implement must not only be incentive compatible in the sense that we have studied so far, but must also satisfy *individual rationality* (or *participation*) *constraints* that assure that each agent i actually wishes to participate in the mechanism. If the expected externality mechanism does not satisfy these constraints, we will need to consider other mechanisms that do. We will have more to say about this issue in Sections 23.E and 23.F, but for now suffice it to say that for this reason, as well as others, we may be interested in identifying *all* of the social choice functions that are Bayesian implementable in this environment.

In the remainder of this section we do this for the special, but often-studied, class of cases in which agents' preferences take a form that is linear in their type, and their types are independently distributed.

Bayesian Incentive Compatibility with Linear Utility

Suppose now that each agent i 's Bernoulli utility function takes the form

$$u_i(x, \theta_i) = \theta_i v_i(k) + (\bar{m}_i + t_i).$$

As before, we shall normalize $\bar{m}_i = 0$ for all i . We also suppose that each agent i 's type lies in an interval $\Theta_i = [\underline{\theta}_i, \bar{\theta}_i] \subset \mathbb{R}$ with $\underline{\theta}_i \neq \bar{\theta}_i$, and that the agents' types are statistically independent. We let the distribution function of θ_i be denoted $\Phi_i(\cdot)$, and we assume that it has an associated density $\phi_i(\cdot)$ satisfying $\phi_i(\theta_i) > 0$ for all $\theta_i \in [\underline{\theta}_i, \bar{\theta}_i]$.

We begin by deriving a necessary and sufficient condition for a social choice function $f(\cdot) = (k(\cdot), t_1(\cdot), \dots, t_I(\cdot))$ to be Bayesian incentive compatible. It is convenient to define $\bar{t}_i(\hat{\theta}_i) = E_{\theta_{-i}}[t_i(\hat{\theta}_i, \theta_{-i})]$; this is agent i 's expected transfer given that he announces his type to be $\hat{\theta}_i$ and that all agents $j \neq i$ truthfully reveal their types. Likewise, we let $\bar{v}_i(\hat{\theta}_i) = E_{\theta_{-i}}[v_i(k(\hat{\theta}_i, \theta_{-i}))]$ denote agent i 's expected "benefit"